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**From DAY 1 to SEM 1 - we got
you covered**



Derivatives

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$$\frac{d(k)}{dx} = 0$$

$$\frac{d(kx)}{dx} = k$$

$$\frac{d(x^n)}{dx} = nx^{n-1}$$

$$\frac{d(x)}{dx} = 1$$

$$\frac{d(\sqrt{x})}{dx} = \frac{1}{2\sqrt{x}}$$

$$\frac{d(\log x)}{dx} = \frac{1}{x}$$

$$\frac{d\left(\frac{1}{x}\right)}{dx} = \frac{-1}{x^2}$$

$$\frac{d(\log_a x)}{dx} = \frac{1}{x \log a}$$

$$\frac{d(a^x)}{dx} = a^x \log a$$

$$\frac{d(e^x)}{dx} = e^x$$

$$\frac{d(\sin x)}{dx} = \cos x$$

$$\frac{d(\cot x)}{dx} = -\operatorname{cosec}^2 x$$

$$\frac{d(\cos x)}{dx} = -\sin x$$

$$\frac{d(\sec x)}{dx} = \sec x \cdot \tan x$$

$$\frac{d(\tan x)}{dx} = \sec^2 x$$

$$\frac{d(\operatorname{cosec} x)}{dx} = -\operatorname{cosec} x \cdot \cot x$$

$$\frac{d(\sin^{-1} x)}{dx} = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{d(\cot^{-1} x)}{dx} = \frac{-1}{1+x^2}$$

$$\frac{d(\cos^{-1} x)}{dx} = \frac{-1}{\sqrt{1-x^2}}$$

$$\frac{d(\sec^{-1} x)}{dx} = \frac{1}{x\sqrt{x^2-1}}$$

$$\frac{d(\tan^{-1} x)}{dx} = \frac{1}{1+x^2}$$

$$\frac{d(\operatorname{cosec}^{-1} x)}{dx} = \frac{-1}{x\sqrt{x^2-1}}$$

Integration

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$$\int c \cdot dx = c$$

$$\int k \cdot dx = (kx + c)$$

$$\int x^n \cdot dx = \frac{x^{n+1}}{n+1} + c \quad \text{----- } n \neq -1$$

$$\int \frac{1}{x} \cdot dx = \log x + c$$

$$\int a^x \cdot dx = \frac{a^x}{\log a} + c$$

$$\int e^x \cdot dx = e^x + c$$

$$\int \sin x \cdot dx = -\cos x + c$$

$$\int \cos x \cdot dx = \sin x + c$$

$$\int \sec^2 x \cdot dx = \tan x + c$$

$$\int \operatorname{cosec}^2 x \cdot dx = -\cot x + c$$

$$\int \sec x \cdot \tan x \cdot dx = \sec x + c$$

$$\int \operatorname{cosec} x \cdot \cot x \cdot dx = -\operatorname{cosec} x + c$$

$$\int \tan x \cdot dx = \log |\sec x| + c$$

$$\int \cot x \cdot dx = \log |\sin x| + c$$

$$\begin{aligned}\int \sec x \cdot dx &= \log |\sec x + \tan x| + c \\ &= \log \left| \tan \left(\frac{\pi}{4} + \frac{x}{2} \right) \right| + c\end{aligned}$$

$$\begin{aligned}\int \operatorname{cosec} x \cdot dx &= \log |\operatorname{cosec} x - \cot x| + c \\ &= \log \left| \tan \frac{x}{2} \right| + c\end{aligned}$$

Single Variable Calculus

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* Mean Value Theorem :

- It is divided into 3 theorems

- 1) Rolle's Theorem.
- 2) Lagrange's Mean Value Theorem.
- 3) Cauchy's Mean Value Theorem.

1) Rolle's Theorem :

Let $f(x)$ be a function defined on closed interval $[a, b]$ s.t. it satisfies 3 conditions :-

i) $f(x)$ is continuous on $[a, b]$
 $a \leq x \leq b$

ii) $f(x)$ is differentiable on (a, b)
 $a < x < b$
 $f(a) = f(b)$

iii) $\exists c \in (a, b)$, s.t. $f'(c) = 0$

NOTE

- 1) Polynomial f^n is continuous & differentiable everywhere.
- 2) Expo. f^n , sine f^n , cos f^n are continuous and differentiable everywhere.
- 3) Logarithmic f^n is continuous and differentiable in its domain $\{0, \infty\}$.
- 4) Modulus f^n is always continuous.

Ex- Verify Rolle's Theorem for the function

$$f(x) = x^2 - 6x + 8 \quad \text{in } [2, 4]$$

→

i) Continuity :

The given function is polynomial

∴ The $f(x)$ is continuous on $[2, 4]$.

ii) Differentiability :

$$f(x) = x^2 - 6x + 8$$

$$f'(x) = 2x - 6$$

∴ The $f(x)$ is diff. on $(2, 4)$

iii) Equal end point :

$$f(2) = f(4)$$

$$\therefore f(2) = (2)^2 - 6(2) + 8 = f(4) = (4)^2 - 6(4) + 8$$

$$f(2) = 0 = f(4) = 0$$

$$\therefore f(2) = f(4)$$

∴ $f(x)$ satisfied all condiⁿ of Rolle's theorem.

∴ $\exists c \in (2, 4)$ s.t.

$$f'(c) = 0$$

$$f'(c) = 2c - 6$$

$$\therefore 0 = 2c - 6$$

$$\therefore \underline{c = 3}$$

$$c = 3 \in (2, 4)$$

Hence, Rolle's theorem verified.

Ex - Verify Rolle's Theorem for the function $f(x) = \sin x + \cos x - 1$ on $[0, \pi/2]$

→ i) continuity :

The given function is trigonometric

$\therefore f(x)$ is continuous on $[0, \pi/2]$

ii) differentiability :

$$f(x) = \sin x + \cos x - 1$$

$$f'(x) = \cos x - \sin x - 0$$

$$\therefore f'(x) = \cos x - \sin x$$

$\therefore f(x)$ is diff. on $(0, \pi/2)$

iii) Equal end point :

$$f(0) = f(\pi/2)$$

$$f(0) = \sin 0 + \cos 0 - 1$$

$$= 0 + 1 - 1$$

$$f(0) = 0 \quad \dots \dots \dots (1)$$

$$f(\pi/2) = \sin(\pi/2) + \cos(\pi/2) - 1$$

$$= 1 + 0 - 1$$

$$f(\pi/2) = 0 \quad \dots \dots \dots (2)$$

$\therefore$ from (1) & (2)

$$f(0) = f(\pi/2)$$

$\therefore$ All condition of Rolle's theorem are satisfied.

$$\exists c \in (0, \pi/2)$$

$$f'(c) = 0$$

$$\therefore \cos c - \sin c = 0$$

$$\cos c = \sin c$$

$$1 = \frac{\sin c}{\cos c}$$

$$\therefore 1 = \tan c$$

$$\therefore c = \tan^{-1} 1$$

$$\therefore c = \pi/4 \in (0, \pi/2)$$

Hence, Rolle's theorem verified.

Ex - Verify Rolle's theorem.

$$f(x) = (x-1)(x-2)(x-3) \text{ in } [0, 3]$$

→

i) The given f' is $(x-1)(x-2)(x-3)$

$\therefore$ The $f(x)$ is polynomial.

$$\text{ii) } (x-1)(x-2)(x-3)$$

$$= (x^2 - 2x - x + 2)(x-3)$$

$$= (x^2 - 3x + 2)(x-3)$$

$$= x^3 - 3x^2 - 3x^2 + 9x + 6$$

$$= x^3 - 6x^2 + 9x + 6$$

$$= x^3 - 6x^2 + 11x - 6$$

$$= x^3 - 6x^2 + 11x - 6$$

$$f(x) = x^3 - 6x^2 + 11x - 6$$

5 ex. of Rolle's theorem

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i) Continuity :

The given f^n is polynomial.

$\therefore$ The $f(x)$ is continuous on $[0, 3]$

ii) Differentiation :

$$f(x) = x^3 - 6x^2 + 11x - 6$$

$$f'(x) = 3x^2 - 12x + 11$$

$\therefore$ The $f(x)$ is diff. on $[0, 3]$

iii) Equal end point :

$$f(0) = f(3)$$

$$\therefore f(0) = 0 - 0 + 0 - 6 = -6$$

$$f(3) = 27 - 36 + 11 = 2$$

$$f(3) = (3-1)(3-2)(3-3) = 0$$

$$\therefore f(0) \neq f(3)$$

$\therefore$ function does not have equal end points $\therefore$

$\therefore$ Rolle's Theorem is not verified.

Ex - Verify Rolle's theorem for function
 $f(x) = x^2 - 10x + 16$ in $[3, 7]$

i) continuity :

The given function is polynomial

$\therefore$ The $f(x)$ is continuous on $[3, 7]$

ii) differentiability :

$$f(x) = x^2 - 10x + 16$$

$$f'(x) = 2x - 10$$

$\therefore$ The $f(x)$ is diff. on $(3, 7)$

iii) Equal end point :

$$f(3) = f(7)$$

$$f(3) = 9 - 30 + 16 = -5$$

$$f(7) = 49 - 70 + 16 = -5$$

$$\therefore f(3) = f(7)$$

$\therefore f(x)$ satisfies all the condition of Rolle's theorem.

$$\exists c \in (3, 7) \text{ s.t. } f'(c) = 0$$

$$\therefore f'(c) = 2c - 10$$

$$0 = 2c - 10$$

$$\therefore 2c = 10$$

$$\underline{c = 5}$$

Ex - Verify Rolle's theorem for function $f(x) = x^2 - 5x + 6$ in $[2, 3]$

i) Continuity :

The given function is polynomial.

$\therefore$ The $f(x)$ is continuous on $[2, 3]$

ii) Differentiability :

$$f(x) = x^2 - 5x + 6$$

$$f'(x) = 2x - 5$$

$\therefore$ The $f(x)$ is diff. on $(2, 3)$

iii) Equal end point :

$$f(2) = f(3)$$

$$f(2) = (2)^2 - 5(2) + 6 = 4 - 10 + 6 = 0$$

$$f(3) = (3)^2 - 5(3) + 6 = 9 - 15 + 6 = 0$$

$$\therefore f(2) = f(3)$$

$f(x)$ satisfies all the condition of Rolle's theorem

$$\exists c \in (2, 3) \quad \text{---} \quad f'(c) = 0$$

$$\therefore f'(c) = 2c - 5$$

$$0 = 2c - 5$$

$$\therefore 2c = 5$$

$$c = \frac{5}{2}$$

$$\underline{\underline{2}}$$

Ex - Verify Rolle's theorem for function
 $\rightarrow f(x) = 3 + (x-2)^{2/3}$ on $[1, 3]$

i) continuity :

The given function is not a polynomial since the power $\frac{2}{3}$ is a rational number.

The $f(x)$ does not follow the first condition.

$\therefore$ Hence, Rolle's theorem is not verified.

Ex - Verify Rolle's theorem for function
 $\rightarrow f(x) = x^2$ on $[-1, 1]$

i) continuity :

The given function is a continuous function.

ii) differentiability :

$f(x) = x^2 \quad \therefore f(x)$ is diff. on $(-1, 1)$

$$f'(x) = 2x$$

iii) Equal end point :

$$f(-1) = f(1)$$

$$f(-1) = (-1)^2 = 1$$

$$f(1) = (1)^2 = 1$$

$$\therefore f(-1) = f(1)$$

$$\exists c \in (-1, 1) \quad f'(c) = 0$$

$$\therefore f'(c) = 2c$$

$$0 = 2c$$

$$\underline{c = 0}$$

Ex- Verify Rolle's theorem for $x^3 - 12x$ in $[0, 2\sqrt{3}]$.

→ i) continuity :

The given function is a polynomial
 $\therefore$ it is a continuous function.

ii) differentiability :

$$f(x) = x^3 - 12x$$

$$\therefore f'(x) = 3x - 12$$

iii) Equal end points :

$$f(0) = (0)^3 - 12(0) = 0$$

$$f(2\sqrt{3}) = (2\sqrt{3})^3 - 12(2\sqrt{3})$$

$$= 8 \cdot 3(\sqrt{3}) - 24\sqrt{3}$$

$$= 24\sqrt{3} - 24\sqrt{3}$$

$$f(2\sqrt{3}) = 0$$

$$\therefore f(0) = f(2\sqrt{3})$$

$f(x)$ follows all condition of Rolle's theorem

$$\therefore \exists c \in (0, 2\sqrt{3}) \quad \text{--- in } f'(c) = 0$$

$$f'(c) = 3x - 12$$

$$0 = 3c - 12$$

$$3c = 12$$

$$c = \frac{12}{3}$$

$$\therefore \underline{c = 4}$$

2) Lanrange's Mean Value Theorem :

i) $f(x)$ is continuous on $[a, b]$.

ii) $f(x)$ is differentiable on (a, b) .

--- $\exists$ atleast one point $c \in (a, b)$ s.t.

$$\boxed{f'(c) = \frac{f(b) - f(a)}{b - a}}$$

Let $f(x)$ be a function defined on $[a, b]$ s.t.

Ex-1 Verify LMVT $f(x) = x^2$ on $[1, 3]$

→ i) continuity :

given $f(x)$ is a polynomial.

$\therefore f(x)$ is a continuous funcⁿ of $[1, 3]$

ii) differentiability :

$$f(x) = x^2$$

$$f'(x) = 2x$$

$\therefore$ given $f(x)$ is differentiable on (a, b) .

$\therefore$ Both condition of LMVT satisfied,

$\therefore \exists$ atleast one point $c \in (a, b)$, s.t.

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$2c = \frac{f(3) - f(1)}{3 - 1} = \frac{(3)^2 - (1)^2}{2}$$

$$2c = \frac{9 - 1}{2}$$

$$\therefore \underline{c = 2} \in (1, 3)$$

Ex-2 Verify LMVT for $f(x) = x(x-2)$ on $[1, 3]$

i) continuity :

$$x(x-2) = x^2 - 2x$$

given $f(x)$ is a polynomial

$\therefore f(x)$ is continuous on $[1, 3]$

ii) differentiability :

$$f(x) = x^2 - 2x$$

$$x^2 - 2x$$

$$f'(x) = 2x - 2$$

$\therefore$ given $f(x)$ is diff. on $(1, 3)$.

both conditions of LMVT are satisfied.

$\therefore \exists$ atleast one point $c \in (1, 3)$ s.t.

$$f'(c) = \frac{f(b) - f(a)}{b - a} = \frac{f(3) - f(1)}{3 - 1}$$

$$2c - 2 = \frac{(3)^2 - 3(2)}{2} - \frac{(1)^2 - 2(1)}{2}$$

$$2c - 2 = \frac{(9 - 6)}{2} - \frac{(1 - 2)}{2}$$

$$2c - 2 = \frac{3 - (-1)}{2} = \frac{4}{2} = 2$$

$$2c - 2 = 2$$

$$2c = 2 + 2$$

$$c = \frac{4}{2}$$

$$\underline{c = 2} \in (1, 3)$$

$\therefore$ LMVT is verified.

Ex. 3 → Verify LMVT for $f(x) = x - 2\sin x$ on $[-\pi, \pi]$

i) continuity :

given f^n is sin function.

$\therefore f(x)$ is continuous on $[-\pi, \pi]$

ii) differentiable :

$$f(x) = x - 2\sin x$$

$$f'(x) = 1 - 2\cos x$$

$$\therefore f'(x) = 1 - 2\cos x$$

$\therefore f(x)$ is diff. on $(-\pi, \pi)$

Both condition of LMVT satisfied.

$\therefore \exists$ atleast one point $c \in (-\pi, \pi)$ st.

$$f'(c) = \frac{f(b) - f(a)}{b - a} = \frac{f(\pi) - f(-\pi)}{\pi - (-\pi)}$$

$$1 - 2\cos c = \frac{f(\pi) - f(-\pi)}{\pi + \pi}$$

$$f(\pi) = \pi - 2\sin \pi = \pi - 2(0) = \pi$$

$$f(-\pi) = -\pi - 2\sin(-\pi) = -\pi - 2(0) = -\pi$$

$$1 - 2\cos c = \frac{\pi - (-\pi)}{\pi + \pi} = \frac{2\pi}{2\pi} = 1$$

$$1 = 1 + 2\cos c \quad \therefore \cos c = 0$$

$$-2\cos c = 0$$

$$2\cos c = 0$$

$$c = \cos^{-1}(0)$$

$$[c = \pi/2] \dots \in (-\pi, \pi)$$

$$\log e = 1$$

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Ex-3 Verify LMVT for $f(x) = e^{0.5x}$ on $[0, 2]$

→

i) continuity :

$f(x)$ is expo. fⁿ

∴ $f(x)$ is continuous on $[0, 2]$

ii) differentiability :

$$f(x) = e^{0.5x}$$

$$f'(x) = e^{0.5x} \cdot 0.5$$

Both condition of LMVT satisfied

∴ $\exists c \in (0, 2)$ s.t.

$$f'(c) = \frac{f(b) - f(a)}{b - a} = \frac{f(2) - f(0)}{2 - 0} = \frac{1}{2}$$

$$e^{0.5c} \cdot 0.5 = \frac{e^{2(0.5)} - e^{0(0.5)}}{2}$$

$$0.5 \cdot e^{0.5c} = \frac{e^1 - e^0}{2} = \frac{e - 1}{2}$$

$$1. e^{0.5c} = e - 1$$

Taking log on both sides

$$\log e^{0.5c} = \log(e - 1)$$

$$0.5c \cdot \log e = \log(e - 1)$$

$$(1) \cdot 0.5c = \log(e - 1)$$

$$c = \frac{1}{0.5} \cdot \log(e-1)$$

$$= 2 \log(e-1)$$

$$= 2 \log(2.72-1) = 2 \log(1.72)$$

$$c = 2(0.2355)$$

$$\therefore \underline{c = 0.4710} \in (0, 2)$$

$\therefore$ LMVT is verified.

Ex-4 Verify LMVT for $f(x) = x^2 - 2x + 4$ on the $[1, 5]$.

→

i) continuity.

The given f^n is polynomial.

$\therefore f(x)$ is the continuous f^n on $[1, 5]$

ii) differentiability.

$$f(x) = x^2 - 2x + 4$$

$$f'(x) = 2x - 2$$

$\therefore f(x)$ is differentiable on $(1, 5)$

Both the condition of LMVT are satisfied

$\therefore \exists c \in (a, b)$ s.t.

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$x^2 - 2x + 4$$

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$$f'(c) = \frac{f(5) - f(1)}{5 - 1}$$

$$2c - 2 = \frac{(5)^2 - 2(5) + 4 - ((1)^2 - 2(1) + 4)}{4}$$

$$2c - 2 = \frac{25 - 10 + 4 - (1 - 2 + 4)}{4}$$

$$= \frac{25 - 6 - 3}{4} = \frac{16}{4} = 4$$

$$2c - 2 = 4$$

$$2c = 4 + 2$$

$$c = \frac{6}{2}$$

$$\therefore \underline{c = 3} \in (1, 5)$$

3) Cauchy's Mean Value Theorem :

$f(x)$ is defined on $[a, b]$, and $g(x)$ is defined on $[a, b]$.

Let $f(x)$ & $g(x)$ be two functions defined on $[a, b]$. s.t. $f(x)$ & $g(x)$

i) $f(x)$ & $g(x)$ are continuous on $[a, b]$

ii) $f(x)$ & $g(x)$ are differentiable on (a, b) .

iii) $g'(x) \neq 0 \quad \exists c \in (a, b)$

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}$$

Ex 1 Verify CMVT for $f(x) = x^2$ & $g(x) = x^3$ on $[1, 2]$.

→

i) continuity :

$$f(x) = x^2, \quad g(x) = x^3$$

both the functions are polynomials.

$\therefore f(x)$ & $g(x)$ are continuous on $[1, 2]$.

ii) differentiability :

$$f(x) = x^2$$

$$g(x) = x^3$$

$$\therefore f'(x) = 2x$$

$$g'(x) = 3x^2$$

$\therefore$ Both $f(x)$ & $g(x)$ are diff. on (a, b) i.e. $(1, 2)$

$$\text{iii) } g'(x) = 3x^2 \neq 0$$

$\therefore$ All the condition of CMVT are satisfied

$\therefore \exists c \in (a, b)$ s.t.

$$f'(c) = \frac{f'(x)}{g'(x)} = \frac{f(b) - f(a)}{g(b) - g(a)}$$

$$f(b) = f(2) = x^2 = (2)^2 = 4$$

$$f(a) = f(1) = x^2 = (1)^2 = 1$$

$$g(b) = x^3 = g(2) = (2)^3 = 8$$

$$g(a) = x^3 = g(1) = (1)^3 = 1$$

$$\therefore f'(c) = \frac{2x}{3x^2} = \frac{4-1}{8-1} = \frac{3}{7}$$

$$\therefore \frac{1}{3c} = \frac{3}{7}$$

$$9c = 14$$

$$c = \frac{14}{9}$$

$$\therefore \underline{\underline{c = \frac{14}{9}}} \in (1, 2)$$

Ex-2 Verify CMVT for $f(x) = \sin x$, $g(x) = \cos x$ in $[0, \pi/2]$.

→ i) continuity,

$$f(x) = \sin x, \quad g(x) = \cos x$$

∴ Both the $f(x)$ & $g(x)$ are $\sin$ & $\cos$ fn.

∴ $f(x)$ & $g(x)$ are continuous on $[0, \pi/2]$

ii) differentiability,

$$f(x) = \sin x$$

$$g(x) = \cos x$$

$$\therefore f'(x) = \cos x$$

$$g'(x) = -\sin x$$

∴ $f(x)$ & $g(x)$ are diff. on $(0, \pi/2)$.

$$\text{iii) } g'(x) = -\sin x \neq 0$$

$$\therefore \exists c \in (0, \pi/2) \text{ s.t.}$$

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}$$

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}$$

$$f(b) = \sin x = \sin \pi/2 = 1$$

$$f(a) = \sin x = \sin 0 = 0$$

$$g(b) = \cos x = \cos \pi/2 = 0$$

$$g(a) = \cos x = \cos 0 = 1$$

$$\therefore \frac{\cos c}{-\sin c} = \frac{1-0}{0-1} = \frac{-1}{1}$$

$$\frac{\cos c}{\sin c} = \frac{1}{1} \quad \text{reciprocal}$$

$$\cot c = 1$$

$$c = \cot^{-1} 1$$

$$c = \frac{\pi}{4}$$

$$\therefore c = \pi/4 \quad \text{on } (0, \pi/2)$$

$\therefore$ CMVT verified.

imp
#

Imp. questions :

Ex - 1

Verify $\exists f(x) = \sin^{-1} x$ where $0 < a < b < 1$.

Use LMVT to prove that

$$\frac{b-a}{\sqrt{1-a^2}} < \sin^{-1} b - \sin^{-1} a < \frac{b-a}{\sqrt{1-b^2}}$$

i) $f(x)$ is continuous on $[0, 1]$

$\therefore \sin^{-1} x = f(x)$, $\sin^{-1}$ is always continuous.

ii) diff.

$$f(x) = \sin^{-1} x$$

$$f'(x) = \frac{1}{\sqrt{1-x^2}}$$

$\therefore f(x)$ is diff. on $(0, 1)$

Both condn of LMVT satisfied.

$\therefore \exists c \in (0, 1)$ s.t.

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$\therefore f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$\frac{1}{\sqrt{1-c^2}} = \frac{\sin^{-1}b - \sin^{-1}a}{b-a} \quad \text{--- (1)}$$

we have give condition $a < b$

$$\therefore a < b$$

we can say that $a < c < b$

$$\therefore a < c < b$$

$$= a^2 < c^2 < b^2 \quad \text{--- Squaring.}$$

$$= -a^2 > -c^2 > -b^2 \quad \text{--- multiplying by -1.}$$

$$= 1 - a^2 > 1 - c^2 > 1 - b^2 \quad \text{--- add +1.}$$

$$= \sqrt{1-a^2} > \sqrt{1-c^2} > \sqrt{1-b^2} \quad \text{--- take square root}$$

$$= \frac{1}{\sqrt{1-a^2}} < \frac{1}{\sqrt{1-c^2}} < \frac{1}{\sqrt{1-b^2}} \quad \text{--- dividing by 1.}$$

from (1) equation.

$$= \frac{1}{\sqrt{1-a^2}} < \frac{\sin^{-1}b - \sin^{-1}a}{b-a} < \frac{1}{\sqrt{1-b^2}}$$

$$= \frac{b-a}{\sqrt{1-a^2}} < \sin^{-1}b - \sin^{-1}a < \frac{b-a}{\sqrt{1-b^2}}$$

$\therefore$ Hence proved. --- multiply by $b-a$

Ex-2 Using LMVT p.t. where $0 < a < b < 1$
 $\frac{b-a}{b} < \log \frac{b}{a} < \frac{b-a}{a}$ $f(x) = \log x$

→

i) continuity :

$\log x$ is continuous on $[0, 1]$.

ii) diff :

$$f(x) = \log x = \frac{1}{x} = f'(x)$$

$f(x)$ is diff. on $(0, 1)$.

Both condition of LMVT satisfied s.t.

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$\frac{1}{c} = \frac{\log b - \log a}{b - a} \quad \text{----- (1)}$$

we have,

$$a < b$$

$$a < c < b$$

$$\frac{1}{a} > \frac{1}{c} > \frac{1}{b} \quad \text{----- dividing by 1}$$

$$\frac{1}{a} > \frac{\log b - \log a}{b - a} > \frac{1}{b} \quad \text{----- from equ (1)}$$

multiplying by $b - a$

$$\frac{b-a}{a} > \log b - \log a > \frac{b-a}{b}$$

$$\frac{b-a}{a} > \log\left(\frac{b}{a}\right) > \frac{b-a}{b}$$

$$\therefore \frac{b-a}{b} < \log\left(\frac{b}{a}\right) < \frac{b-a}{a}$$

Hence Proved.

Ex-3 Prove that

$$\frac{b-a}{1+b^2} < \tan^{-1} b - \tan^{-1} a < \frac{b-a}{1+a^2}$$

Where $0 < a < b < 1$. Hence, show that

$$\frac{\pi}{4} + \frac{3}{25} < \tan^{-1} \frac{4}{3} < \frac{\pi}{4} + \frac{1}{16}$$

→

$$f(x) = \tan^{-1} x$$

i) continuity :

$\therefore \tan^{-1} x$ is continuous on $[0, 1]$

$\therefore f(x)$ is a continuous f^n .

ii) differentiability :

$$f(x) = \tan^{-1} x$$

$$\therefore f'(x) = \frac{1}{1+x^2}$$

$\therefore f(x)$ is differentiable on $(0, 1)$.

Both condition of LMVT satisfied
 $\therefore \exists c \in (0, 1)$ s.t.

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$\frac{1}{1+c^2} = \frac{\tan^{-1}b - \tan^{-1}a}{b-a} \quad \text{--- (1)}$$

we have, $a < b$

$$a < c < b$$

$$a^2 < c^2 < b^2 \quad \text{--- Squaring}$$

$$1+a^2 < 1+c^2 < 1+b^2$$

addition of 1

$$\frac{1}{1+a^2} > \frac{1}{1+c^2} > \frac{1}{1+b^2}$$

reciprocal.

$\therefore$ from equ. (1)

$$\frac{1}{1+a^2} > \frac{\tan^{-1}b - \tan^{-1}a}{b-a} > \frac{1}{1+b^2}$$

multiplying with $b-a$

$$\therefore \frac{b-a}{1+a^2} > \tan^{-1}b - \tan^{-1}a > \frac{b-a}{1+b^2}$$

$$\therefore \frac{b-a}{1+b^2} < \tan^{-1}b - \tan^{-1}a < \frac{b-a}{1+a^2}$$

Hence proved.

$$\frac{b-a}{1+b^2} + \tan^{-1}a < \tan^{-1}b < \tan^{-1}a + \frac{b-a}{1+a^2} \quad \text{--- (2)}$$

To show,

$$\frac{\pi}{4} + \frac{3}{25} < \tan^{-1} \frac{4}{3} < \frac{\pi}{4} + \frac{1}{26}$$

compare / equate above equation with equ. no. (2).

we get

$$\tan^{-1}a = \pi/4$$

$$\tan^{-1}b = \tan^{-1} 4/3$$

$$a = \tan \frac{\pi}{4}$$

$$\therefore b = 4/3$$

$$\therefore \underline{a = 1}$$

put the values of a & b in equ. (2).

$$\therefore \frac{4/3 - 1}{1 + (4/3)^2} + \frac{\pi}{4} < \tan^{-1} \frac{4}{3} < \frac{\pi}{4} + \frac{4/3 - 1}{1 + (1)^2}$$

$$= \frac{1/3}{25/9} + \frac{\pi}{4} < \tan^{-1} \frac{4}{3} < \frac{\pi}{4} + \frac{1/3}{2}$$

$$= \frac{3}{25} + \frac{\pi}{4} < \tan^{-1} \frac{4}{3} < \frac{\pi}{4} + \frac{1}{6}$$

Hence proved / shown.

Ex-4 II Show that,

$$\frac{\pi}{3} - \frac{1}{5\sqrt{3}} > \cos^{-1} \frac{3}{5} > \frac{\pi}{3} - \frac{1}{8}$$

$$f(x) = \cos^{-1} x, \quad 0 < a < b < 1$$

→

i) continuity :

$\cos^{-1} x$ is conti. on $[0, 1]$

$\therefore f(x)$ is a conti. fⁿ

ii) differentiability :

$$f(x) = \cos^{-1} x$$

$$f'(x) = \frac{-1}{\sqrt{1-x^2}}$$

$\therefore f(x)$ is diff. on $(0, 1)$.

Both condition of LMVT satisfied

$\therefore \exists c \in (0, 1)$ s.t.

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$\frac{-1}{\sqrt{1-c^2}} = \frac{\cos^{-1} b - \cos^{-1} a}{b - a}$$

— (1)

we have, $a < b$

$$a < c < b$$

$$a^2 < c^2 < b^2$$

— Squaring

$$-a^2 > -c^2 > -b^2$$

— multiplying by -1

$$\therefore 1 - a^2 > 1 - c^2 > 1 - b^2$$

$$\sqrt{1-a^2} > \sqrt{1-c^2} > \sqrt{1-b^2} \quad \text{— addition of 1}$$

Taking Reciprocal.

taking square root

$$\frac{1}{\sqrt{1-a^2}} < \frac{1}{\sqrt{1-c^2}} < \frac{1}{\sqrt{1-b^2}}$$

$$\frac{-1}{\sqrt{1-a^2}} > \frac{-1}{\sqrt{1-c^2}} > \frac{-1}{\sqrt{1-b^2}} \quad \dots \text{Taking square root}$$

$$\frac{-1}{\sqrt{1-a^2}} > \frac{\cos^{-1}b - \cos^{-1}a}{b-a} > \frac{-1}{\sqrt{1-b^2}}$$

from equ (1)

multiplying by $b-a$

$$\frac{-(b-a)}{\sqrt{1-a^2}} > \cos^{-1}b - \cos^{-1}a > \frac{-(b-a)}{\sqrt{1-b^2}} \quad \text{--- (2)}$$

comparing

$$\cos^{-1}a - \frac{(b-a)}{\sqrt{1-a^2}} > \cos^{-1}b > \cos^{-1}a - \frac{(b-a)}{\sqrt{1-b^2}} \quad \text{--- (2)}$$

comparing equ (2) with given equⁿ.

we get

$$\cos^{-1}a = \pi/3$$

$$\cos^{-1}b = \cos^{-1}3/5$$

$$a = \cos \frac{\pi}{3}$$

$$b = 3/5$$

$$a = 1/2$$

put values of 'a' and 'b' in equ. (2)

$$\frac{\pi}{3} - \frac{(3/5 - 1/2)}{\sqrt{1 - (1/2)^2}} > \cos^{-1} \frac{3}{5} > \frac{\pi}{3} - \frac{(3/5 - 1/2)}{\sqrt{1 - (3/5)^2}}$$

$$\frac{\pi}{3} - \frac{1/10}{\sqrt{3/4}} > \cos^{-1} \frac{3}{5} > \frac{\pi}{3} - \frac{1/10}{\sqrt{16/25}}$$

$$\frac{\pi}{3} - \frac{1/10^5}{\sqrt{3/2}} > \cos^{-1} \frac{3}{5} > \frac{\pi}{3} - \frac{1/10^2}{4/5}$$

$$\therefore \frac{\pi}{3} - \frac{1}{5\sqrt{3}} > \cos^{-1} \frac{3}{5} > \frac{\pi}{3} - \frac{1}{8}$$

Hence shown.

imp.
#Maclaurin's Series :

Any f^n of x i.e. $f(x)$ is expressed in ascending power of x , where $f(x)$ and successive derivative is finite.

$$f(x) = f(0) + f'(0)x + \frac{f''(0)x^2}{2!} + \frac{f'''(0)x^3}{3!} + \dots + \frac{f^n(0)x^n}{n!}$$

Some standard formulae :

$$1) e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$$

$$2) e^{-x} = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \frac{x^4}{4!} - \dots$$

$$3) \sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

$$4) \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

$$5) \tan x = x + \frac{x^3}{3} + \frac{2}{15}x^5 + \frac{17}{315}x^7 + \dots$$

$$6) \sinh x = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^7}{7!} + \dots$$

$$7) \cosh x = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \frac{x^6}{6!} + \dots$$

$$8) \tanh x = x - \frac{x^3}{3} + \frac{2}{15} x^5 - \frac{17}{315} x^7 + \dots$$

$$9) \log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} - \dots$$

$$10) \log(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \frac{x^5}{5} - \dots$$

$$11) (1+x)^n = 1 + nx + \frac{n(n-1)}{2!} x^2 + \frac{n(n-1)(n-2)}{3!} x^3 + \dots$$

$$12) \frac{1}{1+x} = (1+x)^{-1} \\ = 1 - x + x^2 - x^3 + x^4 - x^5 + \dots$$

$$13) \frac{1}{1-x} = (1-x)^{-1} \\ = 1 + x + x^2 + x^3 + x^4 + x^5 + \dots$$

Ex-1 Expand $\sin x$ upto x^7 by using Maclaurin's series.

→

$$f(x) = \sin x$$

$$f(0) = \sin 0 = 0 \quad \therefore f(0) = 0$$

$$f'(x) = \cos x$$

$$f'(0) = \cos 0 = 1 \quad \therefore f'(0) = 1$$

$$f''(x) = -\sin x$$

$$f''(0) = -\sin 0 = 0 \quad \therefore f''(0) = 0$$

$$f'''(x) = -\cos x$$

$$f'''(0) = -\cos(0) = -1 \quad \therefore f'''(0) = -1$$

$$f^{(4)}(x) = \sin x$$

$$f^{(4)}(0) = \sin 0 = 0 \quad \therefore f^{(4)}(0) = 0$$

$$f^{(5)}(x) = \cos x$$

$$f^{(5)}(0) = \cos 0 = 1 \quad \therefore f^{(5)}(0) = 1$$

$$f^{(6)}(x) = -\sin x$$

$$f^{(6)}(0) = -\sin 0 = 0 \quad \therefore f^{(6)}(0) = 0$$

$$f^{(7)}(x) = -\cos x$$

$$f^{(7)}(0) = -\cos 0 = -1 \quad \therefore f^{(7)}(0) = -1$$

$$\therefore \sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

Ex-2 Expand $\tan x$ upto x^5 .

→

$$f(x) = \tan x$$

$$\therefore f(0) = \tan(0) = 0$$

$$\therefore f(0) = 0$$

$$\therefore f'(x) = \sec^2 x$$

$$f'(0) = \sec^2(0) = (1)^2 = 1 \quad \therefore f'(0) = 1$$

$$\begin{aligned} f''(x) &= 2 \sec x \cdot \sec x \cdot \tan x \\ &= 2 \sec^2 x \cdot \tan x \end{aligned}$$

$$\begin{aligned} f''(0) &= 2 \sec^2(0) \cdot \tan(0) \\ &= 0 \end{aligned}$$

$$\therefore f''(0) = 0$$

$$\begin{aligned} f'''(x) &= 2 [\sec^2 x \cdot \sec^2 x + \tan x \cdot 2 \sec^2 x \cdot \tan x] \\ &= 2 [\sec^4 x + 2 \sec^2 x \tan^2 x] \end{aligned}$$

$$f'''(0) = 2 [1 + 0]$$

$$f'''(0) = 2$$

$$\therefore f'''(0) = 2$$

$$\begin{aligned} f^{IV}(x) &= 2 [4 \sec^3 x + 2 \cdot 2 \sec^2 x \cdot \tan x \cdot 2 \tan x \sec^2 x] \\ &= 2 [4 \cdot (1) + 4 \sec^4 x \cdot 2 \tan^2 x] \end{aligned}$$

$$f^{IV}(0) = 2 [4 + 0]$$

$$\therefore f^{IV}(0) = 8$$

$$\therefore f^{IV}(0) = 8$$

$$f^V(x) = 2 [12 \sec^2 x \cdot$$

$$f^{IV}(x) = 4 \sec x \cdot \sec x \cdot \tan x + 8 \sec^2 x \cdot \tan x$$

$$\therefore f^{IV}(0) = 0$$

$$f^V(x) = 4 [\sec^2 x \cdot \sec^2 x + \tan x \cdot 2 \sec x \cdot \sec x \cdot \tan x] + 2 [\sec^4 x \cdot \sec^2 x + \tan x \cdot 4 \sec x \cdot \sec x \cdot \tan x] + 2 [2(\tan^3 x \cdot 2 \sec x \cdot \tan x + \sec^2 x \cdot 3 \tan^2 x \cdot \sec^2 x) + 2(\sec^4 x \cdot \sec^2 x + \tan x \cdot 4 \sec^3 x \cdot \sec x \cdot \tan x)]$$

$$f^V(0) = 4 \sec^4 x + 8 \sec^6 x$$

$$f^V(0) = 4 + 4 \sec^6 x + 8 \sec^6 x$$

$$= 4 + 4(1) + 8(1)$$

$$\therefore f^V(0) = 16$$

$$\therefore \tan x = 0 + 1x + 0 \frac{x^2}{2!} + 2 \frac{x^3}{3!} + 0 \frac{x^4}{4!} +$$

$$16 \frac{x^5}{5!}$$

$$= x + \frac{2x^3}{3!} + 16 \frac{x^5}{5!} + \dots$$

$$= x + \frac{2 \cdot x^3}{6 \cdot 3} + \frac{16 \cdot x^5}{120 \cdot 15} + \dots$$

$$\tan x = x + \frac{x^3}{3} + \frac{2x^5}{15} + \dots$$

Ex - 3Expand $\sec x$ by Maclaurin's Series upto x^4 .

→

$$f(x) = \sec x$$

$$\therefore f(0) = \sec(0) = 1 \quad \therefore f'(0) = 1$$

$$f'(x) = \sec x \cdot \tan x$$

$$f'(0) = \sec(0) \cdot \tan(0) = 0 \quad \therefore f''(0) = 0$$

$$f''(x) = \sec x \cdot \sec^2 x + \tan x \cdot \sec x \cdot \tan x$$

$$= \sec^3 x + \tan^2 x \cdot \sec x$$

$$f''(0) = \sec^3(0) + \tan^2(0) \cdot \sec(0)$$

$$= 1 + (0)$$

$$f''(0) = 1$$

$$\therefore f''(0) = 1$$

$$f'''(x) = (3 \sec^2 x \cdot \sec x \cdot \tan x) + (\tan^2 x \cdot \sec x \cdot \tan x + \sec x \cdot (\sec^2 x)^2) 2 \tan x$$

$$= (3 \sec^3 x \cdot \tan x + \tan^3 x \cdot \sec x + \sec^3 x \cdot 2 \tan x)$$

$$f'''(0) = 3(1) \cdot (0) + (0) \cdot (1) + (1) \cdot 2(0)$$

$$\therefore f'''(0) = 0$$

$$\therefore f'''(0) = 0$$

$$f^{(4)}(x) = 5 [\sec^3 x \cdot \sec^2 x + \tan x \cdot 3 \sec^2 x \cdot \sec x \tan x]$$

$$+ [\tan^3 x \cdot \sec x \tan x + \sec x \cdot 3 \tan^2 x \sec^4 x]$$

$$+ 5 \sec^4 x \cdot \sec x \cdot \tan x$$

$$f^{(4)}(0) = 5 [(1) \cdot (1) + (0) \cdot 3(1) \cdot (1)(0)] +$$

$$[(0) \cdot (1)(0) + (1) \cdot 3(0) \cdot (1)] +$$

$$5 [(1)(1) \cdot (0)]$$

$$f^{IV}(0) = 5 [1+0] + 0 + 0 \\ = 5 [1]$$

$$\therefore f^{IV}(0) = 5$$

$$f(x) = f(0) + f'(0)x + \frac{f''(0)x^2}{2!} + \frac{f'''(0)x^3}{3!} + \frac{f^{IV}(0)x^4}{4!}$$

$$\therefore \sec x = 1 + 0x + \frac{1x^2}{2!} + \frac{0x^3}{3!} + \frac{5x^4}{4!}$$

$$\therefore \sec x = 1 + \frac{x^2}{2} + \frac{5x^4}{24}$$

$$\therefore \sec x = \frac{x^2}{2} + 1 + \frac{x^2}{2} + \frac{5x^4}{24} + \dots$$

Ans:

$$\therefore \sec x = 1 + \frac{x^2}{2} + \frac{5x^4}{24} + \dots$$

Right answer :

$$\sec x = 1 + \frac{x^2}{2} + \frac{5x^4}{24} + \dots$$

$$\frac{du}{dv} = \frac{v u' - uv'}{v^2}$$

$$\frac{du}{x} \cdot \frac{dv}{dx} = uv' + v'u$$

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Ex-4 → Expand $\log(1+e^x)$ upto x^3

$$f(x) = \log(1+e^x) \quad \therefore f(0) = 0$$

$$f(0) = \log(1+e^0)$$

$$= \log 2$$

$$f(0) = 0$$

$$f'(x) = \frac{1 \cdot e^x}{1+e^x} = \frac{e^x}{1+e^x}$$

$$f'(0) = \frac{e^{(0)}}{1+e^{(0)}} = \frac{1}{1+1} = \frac{1}{2} \quad \therefore f'(0) = \frac{1}{2}$$

$$f''(x) = \frac{e^x \cdot e^x - e^x (1+e^x)}{(1+e^x)^2} \quad \text{--- U/V rule}$$

$$=$$

$$= \frac{(1+e^x) e^x - e^x \cdot e^x}{(1+e^x)^2}$$

$$= \frac{e^x + (e^{2x}) - e^{2x}}{(1+e^x)^2} = \frac{e^x}{(1+e^x)^2}$$

$$f''(0) = \frac{e^0}{(1+e^0)^2} = \frac{1}{(1+1)^2} = \frac{1}{4}$$

$$\therefore f''(0) = 1/4$$

$$f'''(x) = \frac{(1+e^x)^2 \cdot e^x - e^x \cdot 2(1+e^x) \cdot e^x}{(1+e^x)^4}$$

$$f'''(x) = \frac{(1+e^x) [(1+e^x) \cdot e^x - e^x \cdot 2e^x]}{(1+e^x)^4}$$

$$= \frac{e^x + e^{2x} - 2e^{2x}}{(1+e^x)^3}$$

$$f'''(x) = \frac{e^x - e^{2x}}{(1+e^x)^3}$$

$$f'''(0) = \frac{e^0 - e^{2(0)}}{(1+e^0)^3} = \frac{1 - 1}{(1+1)^3} = \frac{0}{1} = 0$$

$$\therefore f'''(0) = 0$$

$$\therefore \log(1+e^x) = 0 + \frac{1}{2}x + \frac{1}{4} \frac{x^2}{2!} + \frac{0}{3!}x^3$$

$$= \frac{x}{2} + \frac{x^2}{8} + 0 \dots$$

Ans:

$$\therefore \log(1+e^x) = \frac{x}{2} + \frac{x^2}{8} + \dots$$

Ex-5

Expand $e^{x \cos x}$ and prove that

$$e^{x \cos x} = 1 + x + \frac{x^2}{2} - \frac{x^3}{3} + \dots$$

→

we know that,

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$$

$$\therefore e^{x \cos x} = e^{x \left(1 - \frac{x^2}{2!}\right)}$$

$$= e^{x - \frac{x^3}{2}}$$

$$\therefore = 1 + \left(x - \frac{x^3}{2}\right) + \frac{1}{2} \left(x - \frac{x^3}{2}\right)^2 + \frac{1}{6} \left(x - \frac{x^3}{2}\right)^3$$

$$= 1 + x - \frac{x^3}{2} + \frac{1}{2} x^2 - \frac{x^4}{2} + \frac{x^6}{4} + \frac{1}{6} x^3$$

$$= 1 + x - \frac{x^3}{2} + \frac{1}{2} x^2 + \frac{1}{6} x^3$$

power are more
so do not consider

$$= 1 + x + \frac{x^2}{2} + \frac{x^3}{3} \left(\frac{1}{6} - \frac{1}{2}\right)$$

$$= 1 + x + \frac{x^2}{2} - \frac{x^3}{3}$$

Hence proved.

Ex-6 P.t. $\sqrt{1 + \sin x} = 1 + \frac{x}{2} - \frac{x^2}{8} - \frac{x^3}{48} + \frac{x^4}{384} + \dots$

→ $f(x) = \sqrt{1 + \sin x}$

∴ $f(x) = \sqrt{1 + 2 \cdot \sin x/2 \cdot \cos x/2}$ using half angle formula

∴ $f(x) = \sqrt{(\sin x/2)^2 + (\cos x/2)^2 + 2 \sin x/2 \cos x/2}$

∴ $f(x) = \sqrt{((\sin x/2) + (\cos x/2))^2}$

∴ $f(x) = \sin x/2 + \cos x/2$ ①

We know,

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$$

from ①

$$\left[\frac{x}{2} - \frac{1}{6} \left(\frac{x}{2} \right)^3 \right] + \left[1 - \frac{1}{2} \left(\frac{x}{2} \right)^2 + \frac{1}{24} \left(\frac{x}{2} \right)^4 \right]$$

$$= \frac{x}{2} - \frac{x^3}{48} + 1 - \frac{x^2}{8} + \frac{x^4}{384}$$

$$= 1 + \frac{x}{2} - \frac{x^2}{8} - \frac{x^3}{48} + \frac{x^4}{384}$$

Hence proved.

Ex-7 $\rightarrow$ P.L. $(1+x)^x = 1 + x^2 - \frac{x^3}{2} + \frac{5}{6}x^4 + \dots$

Let $(1+x)^x = y$

taking log on both side

$$\log y = \log (1+x)^x$$

$$\therefore \log y = x \log (1+x)$$

$$e^{\log y} = e^{x \log (1+x)}$$

$$y = e^{x \log (1+x)} \quad \text{--- (1)}$$

We know,

$$\log (1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$$

from (1)

$$y = e^{x \left[x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots \right]}$$

$$= e^{\left[x^2 - \frac{x^3}{2} + \frac{x^4}{3} - \frac{x^5}{4} + \dots \right]}$$

not needed

$$= 1 + \left(x^2 - \frac{x^3}{2} + \frac{x^4}{3} - \frac{x^5}{4} \right) + \left(x^2 - \frac{x^3}{2} + \frac{x^4}{3} \right)^2 \frac{1}{2} +$$

$$\frac{1}{6} \left(x^2 - \frac{x^3}{2} + \frac{x^4}{3} \right)^3 + \left(x^2 - \frac{x^3}{2} + \frac{x^4}{3} \right)^4 \frac{1}{24} + \dots$$

$$(a+b+c)^2 = a^2 + b^2 + c^2 + 2a^2b + 2ab^2 + 2c^2a$$

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$$= 1 + \left(\frac{x^2 - \frac{x^3}{2} + \frac{x^4}{3} \right) + \left(\frac{x^4 - \frac{x^6}{4} + \frac{x^8}{9} \right) - \frac{1}{2} + \dots$$

$$= 1 + \left(\frac{x^2 - \frac{x^3}{2} + \frac{x^4}{3} \right) + \frac{x^4}{2} - \frac{x^6}{8} + \frac{x^8}{18} + \dots$$

$$= 1 + x^2 - \frac{x^3}{2} + \frac{x^4}{3} + \frac{x^4}{2} - \dots$$

$$= 1 + x^2 - \frac{x^3}{2} + \left(\frac{1}{3} + \frac{1}{2} \right) x^4 - \dots$$

$$= 1 + x^2 - \frac{x^3}{2} + \frac{5}{6} x^4 + \dots$$

$$= 1 + x^2 - \frac{x^3}{2} + \frac{5}{6} x^4 + \dots$$

Hence Proved.

Ex-8 P.t. $\tan^{-1}x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots$

→ $y = \tan^{-1}x$
diff. w.r.t x .

$$\frac{dy}{dx} = \frac{1}{1+x^2} \quad \text{--- (1)}$$

We have,

$$\frac{1}{1+x} = 1 - x + x^2 - x^3 + x^4 - x^5 + x^6 - x^7 + \dots$$

$$\therefore \frac{dy}{dx} = 1 - x^2 + (x^2)^2 - (x^2)^3 + (x^2)^4 - (x^2)^5 + \dots$$

Squaring x b'coz we have $1/(1+x^2)$ term.

$$\frac{dy}{dx} = 1 - x^2 + x^4 - x^6 + x^8 - x^{10} + \dots$$

Integrating on limit of $0 \rightarrow x$

$$\int_0^x \frac{dy}{dx} = \int_0^x (1 - x^2 + x^4 - x^6 + x^8 - x^{10} + \dots)$$

$$[y]_0^x = \left[x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots \right]_0^x$$

$$\tan^{-1}x - \tan^{-1}0 = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots$$

$$\therefore \tan^{-1}x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots$$

Hence proved.

Ex - 9 $f(x) = \log(\sec x)$ using Maclaurine Series

$$y = \log(\sec x)$$

$$\frac{dy}{dx} = \frac{1}{\sec x} \cdot \sec x \cdot \tan x \quad \text{--- diff wrt } x$$

$$\therefore \frac{dy}{dx} = \tan x \quad \text{--- (1)}$$

We know,

$$\tan x = x + \frac{x^3}{3} + \frac{2}{15}x^5 + \frac{17}{315}x^7 + \dots$$

$$\therefore \frac{dy}{dx} = x + \frac{x^3}{3} + \frac{2}{15}x^5 + \frac{17}{315}x^7 + \dots$$

integrating on limit $0 \rightarrow x$

$$\int_0^x \frac{dy}{dx} = \int_0^x \left[x + \frac{x^3}{3} + \frac{2}{15}x^5 + \frac{17}{315}x^7 + \dots \right]$$

$$[y]_0^x = \left[\frac{x^2}{2} + \frac{x^4}{4} + \frac{2}{15} \frac{x^6}{6} + \frac{17}{315} \frac{x^8}{8} + \dots \right]_0^x$$

$$[y]_0^x = \left[\frac{x^2}{2} + \frac{x^4}{4} + \frac{2}{15} \frac{x^6}{6} + \dots \right]_0^x$$

$$\log \sec x = \frac{x^2}{2} + \frac{x^4}{4} + \frac{x^6}{45} + \dots$$

Example by Substitution method:

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By Substitution we can convert complicated i.e. composite form into standard form.

Some standard expansions:

$$1) \sin^{-1} x = x + \frac{1}{2} \frac{x^3}{3} + \frac{1}{2} \frac{3}{4} \frac{x^5}{5} + \frac{1}{2} \frac{3}{4} \frac{5}{6} \frac{x^7}{7} + \dots$$

$$2) \cos^{-1} x = \frac{\pi}{2} - \sin^{-1} x$$
$$= \frac{\pi}{2} - \left[x + \frac{1}{2} \frac{x^3}{3} + \frac{1}{2} \frac{3}{4} \frac{x^5}{5} + \dots \right]$$

$$3) \tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots$$

$$4) \sinh^{-1} x = x - \frac{1}{2} \frac{x^3}{3} + \frac{1}{2} \frac{3}{4} \frac{x^5}{5} - \frac{1}{2} \frac{3}{4} \frac{5}{6} \frac{x^7}{7} + \dots$$

$$5) \tanh^{-1} x = x + \frac{x^3}{3} + \frac{x^5}{5} + \frac{x^7}{7} + \dots$$

Supportive Trigonometry formulae :

1) $\sin^{-1}(\sin \theta) = \theta$

2) $\sin^{-1}(\cos \theta) = \sin^{-1}\left(\sin \frac{\pi}{2} - \theta\right) = \frac{\pi}{2} - \theta$

	Formula	Form	Substitution
1)	$\sin^2 \theta + \cos^2 \theta = 1$	$\sqrt{1-x^2}$	put $x = \sin \theta$ or $\cos \theta$
2)	$1 + \tan^2 \theta = \sec^2 \theta$ $1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$	$\sqrt{1+x^2}$	put $x = \tan \theta$ or $\cot \theta$
3)	$\sin 2\theta = 2 \sin \theta \cos \theta$ $= \frac{2 \tan \theta}{1 + \tan^2 \theta}$	$2x\sqrt{1-x^2}$ $\frac{2x}{1+x^2}$	put $x = \sin \theta$ or $\cos \theta$ put $x = \tan \theta$
4)	$\cos 2\theta = \cos^2 \theta - \sin^2 \theta$ $= 2\cos^2 \theta - 1$ $= 1 - 2\sin^2 \theta$ $= \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta}$	$2x^2 - 1$ $1 - 2x^2$ $\frac{1-x^2}{1+x^2}$	put $x = \cos \theta$ put $x = \sin \theta$ put $x = \tan \theta$
5)	$\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$	$\frac{2x}{1-x^2}$	put $x = \tan \theta$
6)	$\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$	$3x - 4x^3$	put $x = \sin \theta$

$$7) \cos 3\theta = \frac{4 \cos^3 \theta - 3 \cos \theta}{1}$$

$$4x^3 - 3x$$

$$\text{put } x = \cos \theta$$

$$8) \tan 3\theta = \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta}$$

$$\frac{3x - x^3}{1 - 3x^2}$$

$$\text{put } x = \tan \theta$$

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Ex - 1 Expand $\sin^{-1}\left(\frac{2x}{1+x^2}\right)$ in ascending power of x

$$\rightarrow y = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$$

$$\text{put } x = \tan\theta \quad \therefore \theta = \tan^{-1}x$$

$$y = \sin^{-1}\left(\frac{2\tan\theta}{1+\tan^2\theta}\right)$$

$$\therefore y = \sin^{-1}(\sin 2\theta)$$

$$y = 2\theta$$

$$\therefore y = 2 \cdot (\tan^{-1}x)$$

$$y = 2 \cdot \left[x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots \right]$$

Ex - 2 Expand $\tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right)$ & p.t. $= \frac{1}{2} \left[x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} \right]$

$\rightarrow$

$$y = \tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right)$$

$$\text{put } x = \tan\theta \quad \therefore \theta = \tan^{-1}x$$

$$y = \tan^{-1}\left(\frac{\sqrt{1+\tan^2\theta}-1}{\tan\theta}\right)$$

$$= \tan^{-1}\left(\frac{\sqrt{\sec^2\theta}-1}{\tan\theta}\right)$$

$$= \tan^{-1}\left(\frac{1/\cos\theta - 1}{\sin\theta/\cos\theta}\right)$$

$$= \tan^{-1} \left(\frac{1 - \cos \theta / \cos \theta}{\sin \theta / \cos \theta} \right)$$

$$y = \tan^{-1} \left(\frac{1 - \cos \theta}{\sin \theta} \right)$$

$$y = \tan^{-1} \left(\frac{2 \sin^2 \theta/2}{2 \sin(\theta/2) \cdot \cos(\theta/2)} \right)$$

$$= \tan^{-1} \left(\frac{\sin \theta/2}{\cos \theta/2} \right)$$

$$= \tan^{-1} (\tan \theta/2)$$

$$y = \theta/2$$

$$y = \tan^{-1} x \quad \frac{1}{2}$$

$$y = \frac{1}{2} \left[\frac{x}{3} - \frac{x^3}{5} + \frac{x^5}{7} - \frac{x^7}{9} + \dots \right]$$

Hence proved.

Ex-3 Expand $\cos^{-1} \left(\frac{x - x^{-1}}{x + x^{-1}} \right)$

$$\Rightarrow y = \cos^{-1} \left(\frac{x - x^{-1}}{x + x^{-1}} \right)$$

$$= \cos^{-1} \left(\frac{x - 1/x}{x + 1/x} \right)$$

$$= \cos^{-1} \left(\frac{x^2 - 1/x}{x^2 + 1/x} \right)$$

$$= \cos^{-1} \left(\frac{x^2 - 1}{x^2 + 1} \right)$$

put $x = \tan \theta \quad \therefore \theta = \tan^{-1} x$

$$= \cos^{-1} \left(\frac{\tan^2 \theta - 1}{1 + \tan^2 \theta} \right)$$

$$= \cos^{-1} \left(- \left(\frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} \right) \right)$$

$$= \cos^{-1} (-\cos 2\theta)$$

$$= \cos^{-1} (-\cos 2\theta) = \cos^{-1} (\cos (\pi - 2\theta))$$

$$y = \pi - 2\theta$$

$$y = \pi - 2(\tan^{-1} x)$$

Ans:

$$y = \pi - 2 \left[x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots \right]$$

Ex-4 Expand $\cos^{-1} (4x^3 - 3x)$

$$y = \cos^{-1} (4x^3 - 3x)$$

put $x = \cos \theta \quad \therefore \theta = \cos^{-1} x$

$$y = \cos^{-1} (4\cos^3 \theta - 3\cos \theta)$$

$$= \cos^{-1} (\cos 3\theta)$$

$$y = 3\theta$$

$$y = 3 (\cos^{-1} x)$$

Ans:

$$\therefore y = 3 \left[\frac{\pi}{2} - \left(x + \frac{1}{2} \frac{x^3}{3} + \frac{1}{2} \frac{3}{4} \frac{x^5}{5} + \dots \right) \right]$$

Taylor's Theorem:

Let $f(a+h)$ be a function of h which can be expanded into power of h and, let the expansion of differentiable term any no. of times w.r.t h then,

$$\begin{aligned} f(x) &= f(a+h) \\ &= f(a) + h f'(a) + \frac{h^2}{2!} f''(a) + \frac{h^3}{3!} f'''(a) + \\ &\quad \dots + \frac{h^n}{n!} f^n(a) \end{aligned}$$

Ex-1 Expand $2x^3 + 7x^2 + x - 6$ in power of $(x-2)$

→

$$f(a+h) = f(a) + h f'(a) + \frac{h^2}{2!} f''(a) + \frac{h^3}{3!} f'''(a) \quad \text{--- (1)}$$

$$f(x) = f(a+h)$$

$$f(x) = f(2 + (x-2))$$

$$\therefore h = (x-2) \quad a = 2$$

$$f(a+h) = f(x-2+2) \\ = f(2)$$

$$f(x) = 2x^3 + 7x^2 + x - 6 \rightarrow (x-2)$$

$$\therefore f(a) = 2(a)^3 + 7(a)^2 + a - 6$$

$$\therefore f(2) = 2(2)^3 + 7(2)^2 + 2 - 6$$

$$f(2) = 40$$

$$\therefore f'(a) = 6a^2 + 14a + 1 \\ = 6a^2 + 14a + 1$$

$$f'(2) = 6(2)^2 + 14(2) + 1$$

$$f'(2) = 53$$

$$\therefore f''(a) = 12a + 14$$

$$f''(2) = 12(2) + 14$$

$$f''(2) = 38$$

$$\therefore f'''(a) = 12$$

$$f'''(2) = 12$$

* Equaⁿ ② became.

$$f(x-2+2) = f(2) + (x-2)f'(2) + \frac{(x-2)^2}{2!}f''(2) +$$

$$\frac{(x-2)^3}{3!}f'''(2)$$

$$f(x-2+2) = 40 + (x-2) 53 + \frac{(x-2)^2 (38)}{2} + \frac{(x-2)^3 (12)}{6}$$

$$\therefore f(x-2+2) = 40 + 53(x-2) + 19(x-2)^2 + 2(x-2)^3$$

Ex-2 Expand $x^3 - 2x^2 + 3x + 1$ in power of $(x-1)$.

→ $h = (x-1)$

$$\therefore f(x) = f(a+h) = f(x-1+1)$$

$$\therefore \underline{a=1} \quad \underline{h=(x-1)}$$

$$\therefore f(x) = x^3 - 2x^2 + 3x + 1$$

$$\therefore f(a) = a^3 - 2a^2 + 3a + 1$$

$$\therefore f(a) = 1^3 - 2(1)^2 + 3(1) + 1$$

$$= 1 - 2 + 3 + 1$$

$$f(a) = 3$$

$$\therefore f'(a) = 3a^2 - 4a + 3$$

$$f'(1) = 3(1)^2 - 4(1) + 3$$

$$= 3 - 4 + 3$$

$$f'(a) = 2$$

$$\therefore f''(a) = 6a - 4$$

$$= 6(1) - 4$$

$$f''(a) = 2$$

$$\therefore f'''(a) = 6$$

now,

$$f(x-1+1) = f(1) + f'(1)(x-1) + \frac{f''(1)(x-1)^2}{2!} + \frac{f'''(1)(x-1)^3}{3!}$$

Ans.

$$\therefore f(x-1+1) = 3 + 2(x-1) + \frac{(x-1)^2}{2} + \frac{(x-1)^3}{6}$$

Ex - 3

→

Expand $x^3 - 2x^2 + 3x + 1$ in power of $(x+3)$
 $h = (x+3)$.

$$\therefore f(x) = f(a+h) = f(x+3-3)$$

$$\therefore \underline{a = -3} \quad \underline{h = (x+3)}$$

$$f(x) = x^3 - 2x^2 + 3x + 1$$

$$f(a) = (-3)^3 - 2(-3)^2 + 3(-3) + 1$$

$$= -27 - 18 - 9 + 1$$

$$= -27 - 27 + 1$$

$$f(a) = -53$$

$$f'(a) = 3x^2 - 4x + 3x + 1$$

$$= 3x^2 - 4x + 3$$

$$\therefore f'(-3) = 3(-3)^2 - 4(-3) + 3$$

$$= 3 \cdot 9 + 12 + 3$$

$$= 42 = 42$$

$$\therefore f''(a) = 6x - 4$$

$$f''(-3) = 6(-3) - 4$$

$$= -18 - 4$$

$$= -22$$

$$f'''(a) = 6$$

$$f'''(-3) = 6$$

$$\therefore f(x+3-3) = f(-3) + f'(-3)(x+3) + \frac{f''(-3)}{2!}(x+3)^2 + \frac{f'''(-3)}{3!}(x+3)^3$$

$$\therefore f(x+3-3) = -53 + 42(x+3) - \frac{22}{2}(x+3)^2 + \frac{6}{3!}(x+3)^3$$

$$f(x+3-3) = -53 + 42(x+3) - 11(x+3)^2 + (x+3)^3$$

Indeterminate Form :

$$\frac{0}{0}, \frac{\infty}{\infty}, 0^0, 0^\infty, 1^\infty, \infty - \infty, 0 \times \infty$$

A value of function at $x=a$, which reduces to $\frac{0}{0}, \frac{\infty}{\infty}, 0 \times \infty,$

$\infty - \infty, 0^0, \infty^0, 1^\infty$. This is the indeterminate form.

L Hospital Rule :

Used to evaluate indeterminate form where limit tends to $\frac{0}{0}$ or $\frac{\infty}{\infty}$.

To evaluate the above limit, we take separate derivative of numerator & Denominator
i.e.

$$\lim_{x \rightarrow 0} \frac{d/dx f(x)}{d/dx g(x)} = \frac{f'(x)}{g'(x)}$$

This process is continued until we get a finite limit.

Some important limits :

$$1) \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$2) \lim_{x \rightarrow 0} \frac{x}{\sin x} = 1$$

$$3) \lim_{x \rightarrow 0} \frac{\tan x}{x} = 1$$

$$4) \lim_{x \rightarrow 0} \frac{x}{\tan x} = 1$$

$$5) \lim_{x \rightarrow 0} \frac{\sin^{-1} x}{x} = 1$$

$$6) \lim_{x \rightarrow 0} \frac{x}{\sin^{-1} x} = 1$$

$$7) \lim_{x \rightarrow 0} (1+x)^{1/x} = e$$

$$8) \lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log a$$

$$9) \lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$$

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Ex - 1 $\lim_{x \rightarrow 0} \frac{e^{ax} - e^{-ax}}{\log(1+bx)}$. Evaluate.

→

$$L = \lim_{x \rightarrow 0} \frac{e^{ax} - e^{-ax}}{\log(1+bx)}$$

∴ put $x \rightarrow 0$.

$$\therefore L = \frac{e^{a(0)} - e^{-a(0)}}{\log(1+b(0))} = \frac{e^0 - e^{-0}}{\log(1+0)}$$

$$L = \frac{1-1}{0} = \frac{0}{0} \Rightarrow \text{indeterminate form.}$$

∴ Apply L-Hospital Rule.

$$L = \lim_{x \rightarrow 0} \frac{\frac{d}{dx} (e^{ax} - e^{-ax})}{\frac{d}{dx} \log(1+bx)}$$

$$= \frac{ae^{ax} + ae^{-ax}}{\frac{b}{(1+bx)}}$$

put $x \rightarrow 0$

$$\therefore L = \frac{ae^{a(0)} + ae^{-a(0)}}{b/(1+b(0))} = \frac{a+a}{b/1}$$

$$\boxed{L = \frac{2a}{b}}$$

$$= \frac{2 + 1 + 0}{2 + 0}$$

$$L = \frac{3}{2}$$

Ex 2 Evaluate $\lim_{x \rightarrow 0} \frac{x e^x - \log(1+x)}{x^2}$

$$\rightarrow L = \lim_{x \rightarrow 0} \frac{x e^x - \log(1+x)}{x^2}$$

$$x = 0 \quad , \quad \frac{0 \cdot e^0 - \log(1+0)}{0^2} = \frac{0 - 0}{0} = \frac{0}{0}$$

Apply L - Hospital rule,

$$L = \frac{x \cdot e^x + e^x \cdot 1 - 1/(1+x)}{2x} = \frac{x e^x + e^x \frac{(1+x) - 1}{(1+x)}}{2x}$$

$$\therefore L = \frac{0 \cdot e^0 + e^0 \cdot 1 - 1/(1+0)}{2(0)} = \frac{0 + 1 - 1/1}{0} = \frac{0}{0}$$

Apply L - Hospital rule,

$$L = \frac{x \cdot e^x + e^x \cdot 1 + e^x \cdot x^2 + x e^x - 1}{(1+x)^2}$$

$$= \frac{2(x e^x + e^x) + e^x + x^2 e^x + e^x 2x}{2 + 4x}$$

put

$$x = 0$$

Ex.-3 Find value of a & b s.t.

$$\lim_{x \rightarrow 0} \frac{a \cos x - a + bx^2}{x^4} = \frac{1}{12}$$

→

$$L = \frac{a \cos(0) - a + b(0)^2}{(0)^4} = \frac{1}{12}$$

$$= \frac{a(1) - a + (0)}{0^4} = \frac{a - a + 0}{0} = \frac{0}{0} \text{ form}$$

Apply L Hospital Rule,

$$= \lim_{x \rightarrow 0} \frac{\frac{d}{dx} (a \cos x - a + bx^2)}{\frac{d}{dx} x^4} = \frac{1}{12}$$

$$= \lim_{x \rightarrow 0} \frac{-a \sin x - 0 + 2bx}{4x^3} \quad \text{--- (1)}$$

put $x \rightarrow 0$

$$\lim_{x \rightarrow 0} \frac{-a \sin(0) + 2b(0)}{4(0)^3} = \frac{-a(0) + 0}{(0)} = \frac{0}{0} \text{ form}$$

$$\frac{0}{0} \neq \frac{1}{12}$$

Apply L Hospital rule on eqn (1).

$$= \lim_{x \rightarrow 0} \frac{-a \cos x + 2b}{12x^2} = \frac{1}{12} \quad \text{--- (2)}$$

$$\text{put } x \rightarrow 0 \quad = \frac{-a \cos(0) + 2b}{12(0)} = \frac{-a + 2b}{0}$$

$$\therefore \frac{-a + 2b}{0} = \frac{1}{12}$$

$$\therefore -a + 2b = 0 \quad \text{--- (3)}$$

$$\therefore \boxed{a = 2b}$$

Apply L Hospital Rule on equ. (2).

$$\lim_{x \rightarrow 0} \frac{a \sin x + 0}{24x} = \frac{1}{12} \quad \text{--- (2)}$$

$$\text{put } x \rightarrow 0 = \frac{a \sin(0)}{24(0)} = \frac{0}{0} \neq \frac{1}{12}$$

Apply L Hospital Rule on equ. (4).

$$\lim_{x \rightarrow 0} \frac{a \cos^2 x}{24} = \frac{a \cos(0)}{24} = \frac{a}{24} = \frac{1}{12}$$

$$\text{put } x = 0 \quad \therefore a = 2 \quad b = 1$$

$$\therefore \boxed{a = 2} \quad \boxed{b = 1}$$

Ex. - 4

Trigonometric Reciprocal :

$$1) \sin x = \frac{1}{\operatorname{cosec} x}$$

$$2) \cos x = \frac{1}{\sec x}$$

$$3) \tan x = \frac{1}{\cot x} = \frac{\sin x}{\cos x}$$

$$4) \cot x = \frac{1}{\tan x} = \frac{\cos x}{\sin x}$$

$$5) \sec x = \frac{1}{\cos x}$$

$$6) \operatorname{cosec} x = \frac{1}{\sin x}$$

Algebraic Reciprocal :

$$\frac{a}{b/c} = \frac{ac}{b}$$

$$\frac{a/b}{c} = \frac{a}{bc}$$

$$\frac{a/b}{c/d} = \frac{a}{b} \times \frac{d}{c}$$

$$\log 0 = -\infty$$

$$\log 1 = 0$$

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Exponential Reciprocal

$$1) \quad 1/e^x = e^{-x}$$

$$2) \quad 1/e^{-x} = e^x$$

NOTE: if we get $\frac{\infty}{\infty}$ form then apply

L Hospital Rule only once untill it we convert it 0/0 form by using Reciprocal method.

Ex-1 Evaluate $\lim_{x \rightarrow 0} \frac{\log(\sin 2x)}{\log(\sin x)}$

→

$$\text{Let } L = \lim_{x \rightarrow 0} \frac{\log(\sin 2x)}{\log(\sin x)}$$

$$\text{put } x \rightarrow 0 \quad \frac{\log(\sin 2(0))}{\log(\sin(0))} = \frac{\log 0}{\log 0} = \frac{+\infty}{+\infty} \text{ form}$$

Apply L Hospital Rule,

$$\lim_{x \rightarrow 0} \frac{\cos 2x \cdot 2}{\sin 2x \cdot 2} = \frac{2 \cdot \cot 2x}{\cot x}$$

$$\frac{\cos x}{\sin x}$$

$$2 \lim_{x \rightarrow 0} \frac{\cot 2x}{\cot x} = \frac{1}{\tan 2x} \cdot \frac{\tan x}{1}$$

$$2 \lim_{x \rightarrow 0} \frac{\tan x}{\tan 2x}$$

$$\therefore \text{put } x \rightarrow 0 \quad \neq \frac{2 \cdot \tan(0)}{\tan 2(0)} = \frac{2 \cdot 0}{0} = \text{form}$$

Apply L Hospital Rule.

$$\lim_{x \rightarrow 0} = \frac{2 \cdot \sec^2 x}{\sec^2 2x \cdot 2} = \frac{\sec^2 x}{\sec^2 2x}$$

$$\therefore \text{put } x \rightarrow 0 = \frac{\sec^2(0)}{\sec^2 2(0)} = \frac{1}{1}$$

$$\therefore \boxed{L = 1}$$

Ex - 2

Evaluate $\lim_{x \rightarrow 0} \frac{\log \tan x}{\log x}$

→

$$\text{Let } L = \lim_{x \rightarrow 0} \frac{\log \tan x}{\log x}$$

$$\text{put } x \rightarrow 0$$

$$\therefore L = \frac{\log \tan(0)}{\log(0)} = \frac{\log 0}{\log 0} = \frac{-\infty}{-\infty} = \frac{\infty}{\infty} \text{ form}$$

Apply L Hospital Rule.

$$\therefore \lim_{x \rightarrow 0} \frac{\sec^2 x}{\tan x} = \frac{x \cdot \sec^2 x}{\tan x}$$

$$\frac{1/x}{1/x}$$

put $x = 0$ $\frac{0 \cdot \sec^2(0)}{\tan(0)} = \frac{0}{0}$

Apply $\frac{0}{0}$ Using Reciprocal method.

$$L = \lim_{x \rightarrow 0} \frac{x \cdot 1}{\cos^2 x} \cdot \frac{\cos x}{\sin x}$$

$$L = \lim_{x \rightarrow 0} \frac{x}{\cos x \cdot \sin x}$$

$x = 0$ $\frac{0}{\cos(0) \cdot \sin(0)} = \frac{0}{0}$ form

Apply L Hospital Rule,

$$\therefore \lim_{x \rightarrow 0} \frac{1}{-\sin x \cdot \sin x + \cos x \cdot \cos x}$$

$$\lim_{x \rightarrow 0} \frac{1}{\cos x \cdot \cos x - \sin x \cdot \sin x}$$

$$\lim_{x \rightarrow 0} \frac{1}{\cos^2 x - \sin^2 x}$$

put $x = 0$ $\frac{1}{\cos^2 0 - \sin^2 0} = \frac{1}{1 - 0} = \frac{1}{1}$

$$\therefore \boxed{L = 1}$$

$$\begin{aligned}\log 0 &= -\infty \\ \log \infty &= \infty \\ \log 1 &= 0 \\ \log 0 &= \infty\end{aligned}$$

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Type 3 : $(0 \times \infty)$ form or $(\infty - \infty)$ form

i) $(0 \cdot \infty)$ or $(\infty - \infty)$ can be converted to $0/0$ form by replacing the ∞ term in denominator.

i.e. $\frac{0}{1/\infty}$ i.e. $\frac{0}{0}$ form

ii) If ∞ is obtained due to a logarithmic term, then do not bring it to denominator, since it is difficult to differentiate $\log x$ function, in such a case bring the x^n other than $\log x$ to denominator.

Ex.-1 Evaluate $\lim_{x \rightarrow 0} (x \cdot \log x)$

$\rightarrow L = \lim_{x \rightarrow 0} (x \cdot \log x)$

put $x = 0$

$L = (0 \cdot \log 0)$

$L = 0 \cdot (-\infty)$

$L = (0 \times -\infty)$ form.

$\therefore L = \lim_{x \rightarrow 0} \frac{\log x}{1/x}$

Apply L - Hospital Rule.

$L = \lim_{x \rightarrow 0} \frac{d/dx \log x}{d/dx 1/x}$

$$L = \frac{1}{x} \cdot \frac{-x^2}{1} = (L = -x)$$

$$\therefore x = 0 \text{ put}$$

$$\therefore L = -0$$

$$\therefore \boxed{L = 0}$$

Ex-2 $\lim_{x \rightarrow 0} \sin x \cdot \log x$. Evaluate.

$$\rightarrow L = \lim_{x \rightarrow 0} \sin x \cdot \log x$$

$$L = \sin(0) \cdot \log(0) \quad \text{--- put } x = 0$$

$$L = (0 \times -\infty) \text{ form}$$

$$\therefore \text{Appty } L = \frac{\log x}{1/\sin x} \quad \text{put } x = 0 \quad \frac{-\infty}{\infty}$$

$\therefore$ apply L-Hospital Rule,

$$\therefore L = \lim_{x \rightarrow 0} \frac{(\log x) d/dx}{d/dx (1/\sin x)}$$

$$L = \lim_{x \rightarrow 0} \frac{1/x}{d/dx (\operatorname{cosec} x)}$$

$$L = \frac{1/x}{-\operatorname{cosec} x \cdot \cot x}$$

$$L = -\frac{1}{x} \cdot \frac{1}{\operatorname{cosec} x \cdot \cot x}$$

$$\text{put } x = 0$$

$$L = \frac{-1}{0 \cdot \operatorname{cosec}(0) \cdot \cot(0)}$$

$$L = \frac{+\infty}{-\infty \cdot -\operatorname{cosec}(0) \cdot \cot(0)} = \frac{\infty}{\infty} \text{ form}$$

$$\therefore L = \lim_{x \rightarrow 0} \frac{1}{x \cdot \operatorname{cosec} x \cdot \cot x}$$

$$L = \lim_{x \rightarrow 0} \frac{1}{x} \cdot \sin x \cdot \tan x$$

$$L = \frac{\sin(0) \cdot \tan(0)}{0} = \frac{0}{0} \text{ form}$$

Apply L-Hospital Rule,

$$L = \lim_{x \rightarrow 0} \frac{\cos x \cdot \sec x + \sin x \cdot \sec^2 x}{1}$$

$$L = \frac{\cos(0) \cdot \tan(0) + \sin(0) \cdot \sec^2(0)}{1}$$

$$L = \frac{1 \cdot (0) + (0) \cdot (1)}{1} = \frac{0}{1}$$

$$\therefore \boxed{L = 0}$$

Ex-3 $\lim_{x \rightarrow 1} \left[\frac{x}{x-1} - \frac{1}{\log x} \right]$

→ put $x = 1$

$$L = \frac{1}{(1-1)} - \frac{1}{\log 1} = \frac{1}{0} - \frac{1}{0} = (\infty - \infty) \text{ form}$$

$$L = \lim_{x \rightarrow 1} \left(\frac{x \cdot \log x - (x+1)}{\log x (x-1)} \right)$$

$$L = \frac{1 \log 1 - 1 + 1}{\log 1 (1-1)} = \frac{1 \cdot (0) \cdot 0}{0 \cdot 0} = \frac{0}{0} \text{ form}$$

Apply L - Hospital Rule ,

$$L = \lim_{x \rightarrow 1} \left[\frac{d/dx (x \log x - (x+1))}{d/dx (\log x (x-1))} \right]$$

$$L = \lim_{x \rightarrow 1} \frac{x \cdot \frac{1}{x} + \log x (1) - (1)}{(x-1) \cdot \frac{1}{x} + \log x (1)} \quad \text{--- by UV rule}$$

$$L = \lim_{x \rightarrow 1} \frac{1 + \log x - 1}{\frac{(x-1)}{x} + \log x} \quad \text{--- (1)}$$

$$L = \frac{0}{0} \text{ form -- put } x = 1$$

Apply L Hospital rule on eqn. (1)

$$L = \lim_{x \rightarrow 1} \frac{1/x}{x(1) - (x-1)1 + \frac{1}{x}} \quad \text{--- by } \frac{U}{V} \text{ rule}$$

$$L = \lim_{x \rightarrow 1} \frac{1/x}{\frac{x+x^2}{x^3}}$$

$$L = \lim_{x \rightarrow 1} \frac{1}{x} \cdot \frac{x^3}{x+x^2}$$

put $x = 1$ on $\frac{x}{x+x^2}$

$$L = \frac{1}{1+1^2} \quad \therefore \boxed{L = \frac{1}{2}}$$

Type 4 : 0^0 , ∞^0 , 1^∞ Form.

i) Let $L =$ form of the 0^0 , ∞^0 , 1^∞

ii) take log on both side

iii) We get either $(0 \cdot \infty)$ or $(\infty \cdot 0)$ form

iv) Shift suitable fun to denominator

So we get $\frac{0}{0}$ or $\frac{\infty}{\infty}$ form.

v) Then apply L-Hospital rule on f^n

Ex - 1

Evaluate $\lim_{x \rightarrow \pi/2} (\cos x)^{\cos x}$

→

$$L = \lim_{x \rightarrow \pi/2} (\cos x)^{\cos x}$$

put $x = \pi/2$

$$\lim_{x \rightarrow \infty} \left(\frac{1}{2} \right)^{1/x}$$

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$$L = (\cos \pi/2)^{\cos \pi/2}$$

$$L = (0)^0 \text{ form}$$

taking log on both side,

$$\therefore \log L = \lim_{x \rightarrow \pi/2} \log (\cos x)^{\cos x}$$

$$\log L = \lim_{x \rightarrow \pi/2} \cos x (\log \cos x)$$

$$\log L = \cos \pi/2 \cdot \log (\cos \pi/2)$$

$$= (0 \times (-\infty))$$

put $x = \pi/2$

$$\log L = (0 \times (-\infty)) \text{ form}$$

$$\log L = \lim_{x \rightarrow \pi/2} \frac{\log \cos x}{1/\cos x}$$

$$\log L = \frac{\infty}{\infty} \text{ form}$$

— put $x = \pi/2$

$$\log L = \lim_{x \rightarrow \pi/2} \frac{\log \cos x}{\sec x}$$

$$\log L = \frac{\infty}{\infty}$$

— put $x = \pi/2$

Apply L-Hospital Rule,

$$\log L = \lim_{x \rightarrow \pi/2} \frac{d/dx \log \cos x}{d/dx \sec x}$$

$$= \lim_{x \rightarrow \pi/2} \frac{1/\cos x (-\sin x)}{\sec x \cdot \tan x}$$

$$\log L = \lim_{x \rightarrow \pi/2} \frac{-\tan x}{\sec x \cdot \tan x}$$

$$\log L = \lim_{x \rightarrow \pi/2} \frac{-1}{\sec x} = -\cos x$$

put $x = \pi/2$

$$\log L = (-\cos \pi/2)$$

$$\log L = 0$$

taking antilog on both side,

$$e^{\log L} = e^0$$

$$L = e^0$$

$$L = 1$$

Ex 2 $\lim_{x \rightarrow \infty} \left(\frac{1}{x}\right)^{1/x}$

put $x = \infty$

$$L = \lim_{x \rightarrow \infty} \left(\frac{1}{\infty}\right)^{1/\infty} = 0^0 \text{ form.}$$

taking log on both sides,

$$\log L = \lim_{x \rightarrow \infty} \log \left(\frac{1}{x}\right)^{1/x}$$

$$\log L = \lim_{x \rightarrow \infty} \frac{1}{x} \cdot \log \frac{1}{x}$$

$$\text{put } x = \infty$$

$$= \frac{1}{\infty} \log \frac{1}{\infty}$$

$$= 0 \log 1/\infty$$

$$\log L = 0 \times \infty \text{ form.} \quad \dots \because \log 0 = \infty$$

$$\log L = \lim_{x \rightarrow \infty} \frac{\log 1/x}{1/1/x}$$

$$\log L = \lim_{x \rightarrow \infty} \frac{\log 1/x}{x}$$

$$\therefore \text{put } x = \infty = \frac{\infty}{\infty} \text{ form.}$$

Apply L Hospital Rule,

$$\log L = \lim_{x \rightarrow \infty} \frac{d/dx \log 1/x}{d/dx x}$$

$$\log L = \lim_{x \rightarrow \infty} \frac{1}{1/x} \cdot \frac{-1}{x^2}$$

1

$$\log L = \lim_{x \rightarrow \infty} \frac{-1}{x}$$

$$\text{put } x = \infty$$

$$\log L = 0$$

taking antilog,

$$e^{\log L} = e^0$$

$$L = 1$$

$$\therefore \boxed{L = 1}$$

$$e^0 = 1$$

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Fourier Series :

Periodic function : A fⁿ $f(x)$ is said to be periodic fⁿ if it is defined for all real x and if there is some +ve no. T , s.t.

$$[f(x) + T = f(x)] \text{ for all } x \text{ then,}$$

T is a period of $f(x)$

NOTE

i) $\sin x, \cos x, \sec x, \csc x$, are periodic fⁿ with fundamental period 2π .

ii) $\tan x$ & $\cot x$ are periodic fⁿ with fundamental period π .

iii) The constant fⁿ $f(x) = c$ is also a periodic fⁿ.

iv) $\sin n\pi = 0$

$$\cos n\pi = (-1)^n$$

$$\cos (2n+1)\pi = -1$$

$$\cos (2n-1)\pi = -1$$

*

Dirichlet's Condition :

$f(x)$ is defined and single valued fⁿ in the given interval and also, integral over C_1 to C_2 $f(x) dx$ exists

i) $\int_{c_1}^{c_2} f(x)$ exists.

ii) $f(x)$ may have finite no. of discontinuity.

iii) $f(x)$ has finite no. of maxima & minima.

Fourier Series :

Let $f(x)$ be a periodic fn of period $2L$ and it is defined in the interval $c \leq x \leq c+2L$ and satisfied the Dirichlet's conditions then $f(x)$ can be expressed as,

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right]$$

where, a_0, a_n, b_n are called as a fourier Coeff. & it's given by

$$a_0 = \frac{1}{L} \int_c^{c+2L} f(x) \cdot dx$$

$$a_n = \frac{1}{L} \int_c^{c+2L} f(x) \cdot \cos\left(\frac{n\pi x}{L}\right) dx$$

$$b_n = \frac{1}{L} \int_c^{c+2L} f(x) \cdot \sin\left(\frac{n\pi x}{L}\right) dx$$

Bernoulli's Rule :

$$\int U \cdot V = U V_1 - U' V_2 + U'' V_3 - U''' V_4 + \dots$$

Where,

U' = derivative of U

V_1 = integration of V .

$$\bullet \int e^{ax} \sin(bx) dx = \frac{e^{ax}}{a^2 + b^2} (a \sin bx - b \cos bx)$$

$$\bullet \int e^{ax} \cos(bx) dx = \frac{e^{ax}}{a^2 + b^2} (a \cos bx + b \sin bx)$$

Ex.-1 Find the fourier of f^n
 $f(x) = x^2$ where $0 \leq x \leq 2\pi$ &
 $f(x+2\pi) = f(x)$

→

$$f(x) = x^2 \quad 0 \leq x \leq 2\pi$$

$$\therefore c \leq x \leq c+2L$$

$$\therefore \underline{c=0}$$

$$\underline{L=\pi}$$

$$\therefore c+2L = \pi$$

Fourier's series :

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[\frac{a_n \cos\left(\frac{n\pi x}{L}\right)}{1} + \frac{b_n \sin\left(\frac{n\pi x}{L}\right)}{1} \right]$$

$$a_0 = \frac{1}{L} \int_c^{c+2L} f(x) \cdot dx$$

$$= \frac{1}{\pi} \int_0^{2\pi} x^2 \cdot dx$$

$$a_0 = \frac{1}{\pi} \left[\frac{x^3}{3} \right]_0^{2\pi}$$

$$a_0 = \frac{1}{\pi} \frac{(2\pi)^3 - (0)^3}{3}$$

$$= \frac{1}{\pi} \frac{8\pi^3}{3}$$

$$a_0 = \frac{8\pi^2}{3}$$

$$a_n = \frac{1}{L} \int_c^{c+2L} f(x) \cos\left(\frac{n\pi x}{L}\right) \cdot dx$$

$$\therefore a_n = \frac{1}{\pi} \int_0^{2\pi} x^2 \cos\left(\frac{n\pi x}{\pi}\right)$$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} x^2 \cdot \cos(nx)$$

By Bernoulli's rule,

$$a_n = \frac{1}{\pi} \int_0^{2\pi} x^2 \cdot \frac{\cos \sin nx}{n} - 2x \cdot \frac{\cos nx}{n} + \frac{2 \cdot \sin nx}{n}$$

$$= \frac{1}{\pi} \int_0^{2\pi} x^2 \cdot \frac{\sin nx}{n} + 2x \frac{\cos nx}{n} - 2 \frac{\sin nx}{n}$$

$$= \frac{1}{\pi} \left\{ \left[\frac{(2\pi)^2 \cdot \sin 2n\pi}{n} + 2(2\pi) \frac{\cos 2n\pi}{n^2} - \frac{2 \sin 2n\pi}{n^3} \right] - \right.$$

$$\left. \left[\frac{0^2 \cdot \sin 0n}{n} + 2(0) \frac{\cos(0)n}{n^2} - \frac{2 \sin(0)n}{n^3} \right] \right\}$$

$$= \frac{1}{\pi} \left[0 + 4\pi \frac{(-1)^{2n}}{n^2} - 0 - 0 \right]$$

$$= \frac{4}{\pi^2} \quad \therefore \boxed{a_n = \frac{4}{n^2}}$$

$$b_n = \frac{1}{L} \int_0^{L+2L} f(x) \cdot \sin\left(\frac{n\pi x}{L}\right) \cdot dx$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} x^2 \cdot \sin\left(\frac{n\pi x}{\pi}\right) \cdot dx$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} x^2 \cdot \sin nx \cdot dx$$

By Bernoulli's rule,

$$b_n = \frac{1}{\pi} \left[\frac{x^2 \cdot \cos nx}{n} - 2x \left(\frac{\sin nx}{n^2} \right) + 2 \frac{\cos nx}{n^3} \right]_0^{2\pi}$$

$$b_n = \frac{1}{\pi} \left[-\frac{x^2 \cos nx}{n} + \frac{2x \sin nx}{n^2} + \frac{2 \cos nx}{n^3} \right]_0^{2\pi}$$

$$b_n = \frac{1}{\pi} \left[-\frac{(2\pi)^2 \cos 2n\pi}{n} + \frac{4\pi \sin 2n\pi}{n^2} + \frac{2 \cos 2n\pi}{n^3} \right] - \left[0^2 \frac{\cos 0}{n} + 2(0) \cdot \frac{\sin 0}{n^2} + \frac{2 \cos(0)}{n^3} \right]$$

$$b_n = \frac{1}{\pi} \left[-\frac{4\pi^2 \cdot (1)}{n} + \frac{4\pi \cdot (0)}{n^2} + \frac{2 \cdot (1)}{n^3} \right] - \frac{0 + 2}{n^3}$$

$$b_n = -\frac{4\pi}{n} + 0 + \frac{2}{n^3} - \frac{2}{n^3}$$

$$\therefore \boxed{b_n = -\frac{4\pi}{n}}$$

From ①

$$f(x) = \frac{4 \cdot 8\pi^2}{63} + \sum_{n=1}^{\infty} \left[\frac{4 \cos\left(\frac{n\pi x}{\pi}\right)}{n^2} + \frac{-4\pi \sin\left(\frac{n\pi x}{\pi}\right)}{n} \right]$$

Ans:

$$f(x) = \frac{4\pi^2}{3} + \sum_{n=1}^{\infty} \left[\frac{4 \cos(nx)}{n^2} - \frac{4\pi \sin(nx)}{n} \right]$$

Ex - 2 Find the fourier series of f^n
 $f(x) = e^{-x}$ where $0 \leq x \leq 2\pi$ &
 $f(x+2\pi) = f(x)$.

→

$$f(x) = e^{-x}$$

$$\therefore 0 \leq x \leq 2\pi$$

$$c \leq x \leq c+2L$$

$$\therefore c=0 \quad c+2L=2\pi$$

$$L = \pi$$

fourier series:

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right]$$

$$\therefore a_0 = \frac{1}{L} \int_c^{c+2L} f(x) \, dx$$

$$= \frac{1}{\pi} \int_0^{2\pi} e^{-x} \cdot dx$$

$$= \frac{1}{\pi} \left[\frac{e^{-x}}{-1} \right]_0^{2\pi}$$

$$a_0 = \frac{1}{\pi} \left[\frac{e^{-2\pi}}{-1} - \frac{e^{-0}}{-1} \right] = -\frac{e^{-2\pi}}{\pi} + \frac{e^0}{\pi}$$

$$= \frac{1 - e^{-2\pi}}{\pi}$$

$$a_0 = \frac{1 - e^{-2\pi}}{\pi}$$

$$a_n = \frac{1}{L} \int_c^{c+2L} f(x) \cdot \cos\left(\frac{n\pi x}{L}\right) \cdot dx$$

$$= \frac{1}{\pi} \int_0^{2\pi} e^{-x} \cdot \cos\left(\frac{n\pi x}{\pi}\right) \cdot dx$$

$$= \frac{1}{\pi} \int_0^{2\pi} e^{-x} \cdot \cos(nx) \cdot dx$$

$$a_n = \frac{1}{n} \int_0^{2\pi} e^{-x} \cdot \cos(nx) \cdot dx$$

By shortcut formula,

$$e^{+ax} \cos bx = \frac{e^{+ax}}{a^2 + b^2} [a \cos bx + b \sin bx]$$

$$\therefore \underline{a = -1} \quad \underline{b = n}$$

$$\therefore a_n = \frac{e^{-x}}{(-1)^2 + (-n)^2} [(-1) \cos nx + n \sin nx]$$

$$a_n = \frac{1}{n} \int_0^{2\pi} \frac{e^{-x}}{1+n^2} [-\cos nx + n \sin nx] dx$$

$$a_n = \frac{1}{n} \left\{ \left[\frac{e^{-2\pi}}{1+n^2} [-\cos(2\pi n) + n \sin 2\pi n] \right] - \frac{e^0}{1+n^2} [-\cos(0) + n \sin(0)] \right\}$$

$$a_n = \frac{1}{n} \left[\frac{e^{-2\pi}}{1+n^2} (-1 + 0) - \frac{1}{1+n^2} (-1 + 0) \right]$$

$$= \frac{1}{n} \left[-\frac{e^{-2\pi}}{1+n^2} + \frac{1}{1+n^2} \right]$$

$$a_n = \frac{1}{n} \left(\frac{1 - e^{-2\pi}}{1+n^2} \right)$$

$$\therefore \boxed{a_n = \frac{1}{n} \left(\frac{1 - e^{-2\pi}}{1+n^2} \right)}$$

$$b_n = \frac{1}{b} \int_0^{2\pi} f(x) \cdot \frac{\sin bx\pi}{L} dx$$

$$= \frac{1}{\pi} \int_0^{2\pi} e^{-x} \frac{\sin n\pi x}{\pi} dx$$

$$= \frac{1}{\pi} \int_0^{2\pi} e^{-x} \sin(nx) dx$$

By shortcut formula

$$e^x \sin bx = \frac{e^{ax}}{a^2 + b^2} [a \sin bx - b \cos bx]$$

$\therefore$

$$\therefore \underline{a = -1} \quad b = n$$

$$\therefore b_n = \frac{1}{\pi} \left[\frac{e^{-x}}{1+n^2} (-1 \sin nx - n \cos nx) \right]_0^{2\pi}$$

$$= \frac{1}{\pi} \left[\frac{e^{-2\pi}}{1+n^2} (-1 \sin n2\pi - n \cos n2\pi) \right] -$$

$$\left[\frac{e^0}{1+n^2} (-1 \sin(0) - n \cos n(0)) \right]$$

$$= \frac{1}{\pi} \left[\frac{e^{-2\pi}}{1+n^2} (-n) - \frac{(-n)}{1+n^2} \right]$$

$$= \frac{1}{\pi} \left[\frac{-n e^{-2\pi}}{1+n^2} + \frac{n}{1+n^2} \right]$$

$$\boxed{b_n = \frac{n}{\pi} \left[\frac{1 - e^{-2\pi}}{1+n^2} \right]}$$

Fourier's Series :

Ans:

$$e^{-x} = \frac{1 - e^{-2\pi}}{2\pi} + \sum_{n=1}^{\infty} \left[\frac{1}{\pi} \left(\frac{1 - e^{-2\pi}}{1 + n^2} \right) \cos nx + \frac{n}{\pi} \left(\frac{1 - e^{-2\pi}}{1 + n^2} \right) \sin nx \right]$$

TYPE - 2 $-L \leq x \leq L$

- Ex. 3
- i) f^n is even $a_0 \& a_n$, $b_n = 0$
 - ii) f^n is odd $a_0 \& a_n = 0$, find b_n
 - iii) neither even nor odd.
 $a_0, a_n, b_n \Rightarrow$ find.

Type : 2 : $-L \leq x \leq L$

Whenever the f^n is defined in the interval $[-L \leq x \leq L]$, we need to check the given f^n is even, odd, neither even nor odd.

To identify the above case we replace x by $-x$.

If we get, i) $f(-x) = f(x) \Rightarrow$ even f^n

ii) $f(-x) = -f(x) \Rightarrow$ odd f^n

iii) $f(-x) \neq -f(x) \Rightarrow f^n$ is neither even nor odd.

1) Even f^n

$$a_0 = \frac{2}{L} \int_0^L f(x) \cdot dx \quad b_n = 0$$

$$a_n = \frac{2}{L} \int_0^L f(x) \cdot \cos\left(\frac{n\pi x}{L}\right) \cdot dx$$

2) Odd f^n :

$$a_0 = 0$$

$$a_n = 0$$

$$b_n = \frac{2}{L} \int_0^L f(x) \sin \left(\frac{n\pi x}{L} \right) dx$$

3) neither even nor odd,

$$a_0 = \frac{1}{L} \int_{-L}^L f(x) \cdot dx$$

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cdot \cos \left(\frac{n\pi x}{L} \right) \cdot dx$$

$$b_n = \frac{1}{L} \int_{-L}^L f(x) \cdot \sin \left(\frac{n\pi x}{L} \right) dx$$

Ex-1 find the fourier series to represent the f^n

$$f(x) = \pi^2 - x^2 \text{ in interval } -\pi \leq x \leq \pi \text{ \& } f(x+2\pi) = f(x)$$

$$f(x) = \pi^2 - x^2$$

$$f(-x) = \pi^2 - (-x)^2 = \pi^2 - x^2$$

$$\therefore f(-x) = f(x)$$

$\therefore$ The given f^n is even f^n

$$\therefore b_n = 0.$$

$$a_0 = \frac{2}{L} \int_0^L f(x) \cdot dx$$

$$= \frac{2}{\pi} \int_0^{\pi} (\pi^2 - x^2) \cdot dx$$

$$= \frac{2}{\pi} \left[\pi^2 x - \frac{x^3}{3} \right]_0^{\pi}$$

$$= \frac{2}{\pi} \left[\pi^2 \cdot \pi - \frac{\pi^3}{3} - \pi^2 \cdot 0 + \frac{0^3}{3} \right]$$

$$= \frac{2}{\pi} \left[\pi^3 - \frac{\pi^3}{3} - 0 + 0 \right]$$

$$= \frac{2}{\pi} \left[\frac{3\pi^3 - \pi^3}{3} \right] = \frac{2}{\pi} \cdot \frac{2\pi^3}{3}$$

$$a_0 = \frac{4\pi^2}{3}$$

$$a_n = \frac{2}{L} \int_0^L f(x) \cdot \cos\left(\frac{n\pi x}{L}\right) \cdot dx$$

$$= \frac{2}{\pi} \int_0^{\pi} (\pi^2 - x^2) \cdot \cos\left(\frac{n\pi x}{\pi}\right) \cdot dx$$

$$= \frac{2}{\pi} \int_0^{\pi} (\pi^2 - x^2) \cdot \cos nx \cdot dx$$

By Bernoulli's Rule,

$$= \frac{2}{\pi} \left[(\pi^2 - x^2) \frac{\sin nx}{n} - (-2x) \cdot \frac{\cos nx}{n^2} + \frac{-2 - \sin nx}{n^3} \right]_0^{\pi}$$

put upper limit & lower limit

$$a_n = \frac{2}{\pi} \left[(\pi^2 - \pi^2) \cdot \frac{\sin n\pi}{n} - 2\pi \frac{\cos n\pi}{n^2} + 2 \frac{\sin n\pi}{n^3} \right] - \left[(\pi^2 - 0^2) \frac{\sin 0}{n} - 2(0) \frac{\cos 0}{n^2} + 2 \frac{\sin 0}{n^3} \right]$$

$$a_n = \frac{2}{\pi} \left(-2\pi \frac{(-1)^n}{n^2} \right)$$

$$a_n = \frac{-4(-1)^n}{n^2}$$

$$\therefore \boxed{a_n = \frac{-4(-1)^n}{n^2}}$$

$\therefore$ Fourier's Series

$$(\pi^2 - x^2) = \frac{4\pi^2}{6} + \sum_{n=1}^{\infty} \left[\frac{-4(-1)^n}{n^2} \cos \frac{n\pi x}{\pi} + 0 \right]$$

Ans: $(\pi^2 - x^2) = \frac{4\pi^2}{6} + \sum_{n=1}^{\infty} \left[\frac{-4(-1)^n}{n^2} \cos nx \right]$

Ex-2 Find the Fourier Series for $f(x)$.

$$f(x) = x, \quad -\pi \leq x \leq \pi,$$

Ex-3 Find Fourier series, for $f(x)$
 $f(x) = x - x^2$ $-1 \leq x \leq 1$

→

$$\therefore f(-x) = (-x) - (-x)^2$$

$$f(-x) = (-x) - x^2 \quad \text{--- (1)}$$

$$\therefore f(x) \neq f(-x)$$

given f^n is neither even nor odd.

$$\therefore f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right)$$

$$a_0 = \frac{1}{L} \int_{-1}^1 (x - x^2) \cdot dx$$

$$= \frac{1}{1} \int_{-1}^1 (x - x^2) \cdot dx$$

$$= 1 \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_{-1}^1$$

$$= \left(\frac{1}{2} - \frac{1}{3} \right) - \left(\frac{1}{2} - \frac{1}{3} \right)$$

$$= \frac{1}{6} - \frac{5}{6} = \frac{-4}{6} = \frac{-2}{3}$$

$$\therefore a_0 = \frac{-2}{3}$$

$$a_n = \frac{1}{1} \int_{-1}^1 (x - x^2) \cdot \cos\left(\frac{n\pi x}{1}\right) \cdot dx$$

$$= 1 \int_{-1}^1 (x - x^2) \cos(n\pi x) \cdot dx \quad \text{--- By Bernoulli's rule}$$

$$= \left[\frac{x^2}{2} - \frac{x^3}{3} \cdot \frac{\sin n\pi x}{n\pi} - (1 - 2x) \left(\frac{-\cos n\pi x}{n^2 \pi^2} \right) + (-2) \left(\frac{-\sin n\pi x}{n^3 \pi^3} \right) \right]_{-1}^1$$

$$a_n = \left\{ \left[\frac{(1-1) \sin n\pi}{n\pi} + \frac{(1-2) \cos n\pi}{n^2 \pi^2} + \frac{2 \sin n\pi}{n^3 \pi^3} \right] - \left[\frac{(-1 - (-1)^2) - \sin n\pi}{n\pi} + \frac{(1-2)(-1)}{n^2 \pi^2} - \frac{\cos n\pi}{n^3 \pi^3} + 2 \left(\frac{-\sin n\pi}{n^3 \pi^3} \right) \right] \right\}$$

$$a_n = - \frac{(-1)^n}{n^2 \pi^2} - \frac{3(-1)^n}{n^2 \pi^2}$$

$$a_n = \frac{-4(-1)^n}{n^2 \pi^2}$$

$$b_n = \frac{1}{L} \int_0^L f(x) \cdot \sin\left(\frac{n\pi x}{L}\right) \cdot dx$$

$$= \frac{1}{1} \int_0^1 (x - x^2) \cdot \sin\left(\frac{n\pi x}{1}\right) \cdot dx$$

$$= \frac{1}{1} \left[(x - x^2) \cdot \left(-\frac{\cos n\pi x}{n\pi} \right) - (1 - 2x) \left(-\frac{\sin n\pi x}{n^2 \pi^2} \right) + (-2) \left(\frac{\cos n\pi x}{n^3 \pi^3} \right) \right]_0^1$$

$$= \left\{ \left[\frac{(1-1) \left(-\frac{\cos n\pi}{n\pi} \right) - (1-2) \left(-\frac{\sin n\pi}{n^2 \pi^2} \right) - 2 \frac{\cos n\pi}{n^3 \pi^3} \right] - \left[\frac{(-1) - (-1)^2 \left(\frac{\cos n\pi}{n\pi} \right) - (1+2) \left(\frac{\sin n\pi}{n^2 \pi^2} \right) + 2 \frac{\cos n\pi}{n^3 \pi^3} \right] \right\}$$

$$\left[\frac{(-1) - (-1)^2 \left(\frac{\cos n\pi}{n\pi} \right) - (1+2) \left(\frac{\sin n\pi}{n^2 \pi^2} \right) + 2 \frac{\cos n\pi}{n^3 \pi^3} \right]$$

$$b_n = \frac{-2(-1)^n}{n^3\pi^3} - \frac{2(-1)^n}{n\pi} + \frac{2(-1)^n}{n^3\pi^3}$$

$$\therefore b_n = \frac{-2(-1)^n}{n\pi}$$

$\therefore$ Fourier's series

$$(x-x^2) = \frac{-2}{6} + \sum_{n=1}^{\infty} \left[\frac{-4(-1)^n}{n^2\pi^2} (\cos(n\pi x)) + \frac{-2(-1)^n}{n\pi} \sin(n\pi x) \right]$$

Ans:

$$\therefore (x-x^2) = \frac{-2}{6} + \sum_{n=1}^{\infty} \left[\frac{-4(-1)^n}{n^2\pi^2} (\cos(n\pi x)) + \frac{-2(-1)^n}{n\pi} \sin(n\pi x) \right]$$

Ex-4 Find the fouries series

$$f(x) = 4 - x^2 \quad 0 \leq x \leq 2$$

$\rightarrow$

$$f(x) = 4 - x^2 \quad 0 \leq x \leq 2$$

$$c \leq x \leq c+2L$$

$$\therefore \underline{c=0} \quad c+2L=2$$

$$0+2L=2$$

Fourier's series:

$$\underline{L=1}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right]$$

$$\therefore a_0 = \frac{8\pi^2}{3} \quad \frac{1}{L} \int_c^{c+2L} f(x) \cdot dx$$

$$\begin{aligned}
 a_0 &= \frac{1}{1} \int_0^2 (4-x^2) \cdot dx \\
 &= \int_0^2 (4-x^2) \\
 &= \left[4x - \frac{x^3}{3} \right]_0^2
 \end{aligned}$$

$$a_0 = \left(8 - \frac{8}{3} \right) - \left(0 - \frac{0}{3} \right)$$

$$a_0 = \frac{16}{3}$$

$$\begin{aligned}
 a_n &= \frac{1}{1} \int_0^2 (4-x^2) \frac{\cos nx\pi}{1} \cdot dx \\
 &= \int_0^2 (4-x^2) (\cos nx\pi)
 \end{aligned}$$

By Bernoulli's rule.

$$\begin{aligned}
 a_n &= \left\{ \left[\frac{(4-x^2) \sin nx\pi}{n\pi} - (0-2x) \left(\frac{-\cos nx\pi}{n^2\pi^2} \right) + \right. \right. \\
 &\quad \left. \left. (-2) \left(\frac{-\sin nx\pi}{n^3\pi^3} \right) \right]_0^2 \right\}
 \end{aligned}$$

$$\begin{aligned}
 a_n &= \left\{ \left[\frac{(4-4) \cdot \sin 2n\pi}{n\pi} - (-4) \left(\frac{-\cos 2n\pi}{n^2\pi^2} \right) - 2 \left(\frac{-\sin 2n\pi}{n^3\pi^3} \right) \right. \right. \\
 &\quad \left. \left. - \left[(0) \cdot 0 - (0) + (-2) - (0) \right] \right\}
 \end{aligned}$$

$$a_n = \left[\frac{-4}{n^2\pi^2} \right]$$

$$\therefore a_n = \frac{-4}{n^2\pi^2}$$

$$b_n = \frac{1}{1} \int_0^2 (4-x^2) \cdot \sin\left(\frac{n\pi x}{1}\right) \cdot dx.$$

$$= 1 \int_0^2 (4-x^2) \cdot \sin(n\pi x) \cdot dx.$$

— By Bernoulli rule

$$b_n = \left[(4-x^2) \left(\frac{-\cos n\pi x}{n\pi} \right) - (-2x) \left(\frac{-\sin n\pi x}{n\pi^2} \right) + (-2) \left(\frac{+\cos n\pi x}{n^3\pi^3} \right) \right]_0^2$$

$$b_n = \left[(4-4) \left(\frac{-\cos 2n\pi}{n\pi} \right) - (-4) \left(\frac{-\sin 2n\pi}{n^2\pi^2} \right) + (-2) \left(\frac{\cos 2n\pi}{n^3\pi^3} \right) \right] -$$

$$\left[(4) \left(\frac{-\cos 0}{n\pi} \right) - (4) (0) + (-2) \left(\frac{\cos n\pi(0)}{n^3\pi^3} \right) \right]$$

$$b_n = \left[0 - 0 + (-2) \left(\frac{1}{n^3\pi^3} \right) \right] - \left[\frac{-4}{n\pi} - 0 + (-2) \left(\frac{1}{n^3\pi^3} \right) \right]$$

$$b_n = \left[\frac{-2}{n^3\pi^3} \right] - \left[\frac{-4}{n\pi} - \frac{2}{n^3\pi^3} \right]$$

$$= \left[\frac{-2}{n^3\pi^3} + \frac{4}{n\pi} + \frac{2}{n^3\pi^3} \right]$$

$$\therefore b_n = \frac{4}{n\pi}$$

Ans:

$$(4-x^2) = \left[\frac{16}{6} + \sum_{n=1}^{\infty} \left(\frac{-4}{n^2\pi^2} \right) \cos(n\pi x) + \left(\frac{4}{n\pi} \right) \sin(n\pi x) \right]$$

**
#

Half Range Series:

In half range series $f(x)$ expanded arranged $(0 \rightarrow L)$ $(0, L)$ then the period is $2L$.

Half Range Sine Series:

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right]$$

where,

$$a_0 \neq a_n = 0$$

$$b_n = \frac{2}{L} \int_0^L f(x) \cdot \sin\left(\frac{n\pi x}{L}\right) dx$$

Half Range Cosine series:

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{L}\right) \right]$$

where,

$$a_0 = \frac{2}{L} \int_0^L f(x) \cdot dx$$

$$a_n = \frac{2}{L} \int_0^L f(x) \cdot \cos\left(\frac{n\pi x}{L}\right) \cdot dx$$

$$b_n = 0$$

Ex-1 Find Half-Range cosine series for $f(x)$
 $f(x) = x^2$, $0 \leq x \leq \pi$

$\rightarrow$
 $f(x) = x^2$, $c=0$ $L=\pi$

$$a_0 = \frac{2}{\pi} \int_0^{\pi} x^2 \cdot dx$$

$$= \frac{2}{\pi} \left[\frac{x^3}{3} \right]_0^{\pi}$$

$$a_0 = \frac{2}{\pi} \left[\frac{\pi^3}{3} - \frac{0^3}{3} \right]$$

$$a_0 = \frac{2\pi^2}{3}$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} (x^2) \cdot \cos\left(\frac{n\pi x}{\pi}\right) \cdot dx$$

$$= \frac{2}{\pi} \int_0^{\pi} (x^2) \cdot \cos nx \cdot dx$$

By Bernoulli's Rule,

$$= \frac{2}{\pi} \left[x^2 \cdot \frac{\sin nx}{n} - (2x) \left(-\frac{\cos nx}{n^2} \right) + 2 \left(-\frac{\sin nx}{n^2} \right) \right]_0^{\pi}$$

$$= \frac{2}{\pi} \left[\pi^2 \cdot \frac{\sin n\pi}{n} - (2\pi) \left(-\frac{\cos n\pi}{n^2} \right) + 2 \left(-\frac{\sin n\pi}{n^2} \right) - 0 - 0 + 0 \right]$$

$$= \frac{2}{\pi} \left(0 + \frac{2\pi (-1)^n}{n^2} + 0 - 0 \right)$$

$$b_n = \frac{4(-1)^n}{n^2}$$

Ex-2 Find Half Range sine series for f^n
 $f(x) = 5\pi^2 - x^2$, $0 \leq x \leq \pi$

→

$$f(x) = \pi^2 - x^2$$

$$0 \rightarrow L \quad \therefore L = \pi$$

it is a sine series $\therefore a_n \& a_0 = 0$

$$\begin{aligned} \therefore b_n &= \frac{2}{L} \int_0^L f(x) \cdot \sin\left(\frac{n\pi x}{L}\right) \cdot dx \\ &= \frac{2}{\pi} \int_0^{\pi} \pi^2 - x^2 \cdot \sin\left(\frac{n\pi x}{\pi}\right) \cdot dx \\ &= \frac{2}{\pi} \int_0^{\pi} \pi^2 - x^2 \cdot \sin nx \cdot dx \end{aligned}$$

By using Bernoulli's rule.

$$b_n = \frac{2}{\pi} \left[\pi^2 - x^2 \left(\frac{-\cos nx}{n} \right) - (0 - 2x) \left(\frac{-\sin nx}{n^2} \right) + \frac{-2}{n^3} \left(+\cos nx \right) \right]_0^{\pi}$$

$$b_n = \frac{2}{\pi} \left[\pi^2 - x^2 \left(\frac{-\cos nx}{n} \right) - (-2x) \left(\frac{-\sin nx}{n^2} \right) + \frac{(-2)}{n^3} \left(\frac{\cos nx}{n^3} \right) \right]_0^{\pi}$$

$$\begin{aligned} \therefore b_n &= \frac{2}{\pi} \left[(\pi^2 - \pi^2) \left(\frac{-\cos n\pi}{n} \right) - (2\pi) \left(\frac{\sin n\pi}{n^2} \right) + \frac{(-2)}{n^3} \left(\frac{\cos n\pi}{n^3} \right) \right]_0^{\pi} - \left[\pi^2 \left(\frac{-1}{n} \right) - 0 + \frac{-2}{n^3} \right] \end{aligned}$$

$$b_n = \frac{2}{\pi} \left[\frac{-2(-1)^n}{n^3} - \left(\frac{-\pi^2}{n} - \frac{2}{n^3} \right) \right]$$

$$\therefore b_n = \frac{2}{\pi} \left[\frac{-2(-1)^n}{n^3} + \frac{\pi^2}{n} + \frac{2}{n^3} \right]$$

$$b_n = \frac{2}{\pi} \left[\frac{2 - 2(-1)^n}{n^3} + \frac{\pi^2}{n} \right]$$

$$\therefore b_n = \frac{2}{\pi} \left[\frac{2 - 2(-1)^n}{n^3} + \frac{\pi^2}{n} \right]$$

fouries series.

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right]$$

Ans:

$$\pi^2 - x^2 = \sum_{n=1}^{\infty} \left[\frac{2}{\pi} \left(\frac{2 - 2(-1)^n}{n^3} + \frac{\pi^2}{n} \right) \cdot \sin(nx) \right]$$

Ex-3 Find Half Range cosine series,
 $f(x) = (x-x^2)$, $0 \leq x \leq 1$.

→

It is a cosine series,

$$a_0 = \frac{2}{L} \int_0^L f(x) \cdot dx$$

$$= \frac{2}{1} \int_0^1 (x-x^2) dx$$

$$= 2 \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1$$

$$= 2 \left[\left(\frac{1}{2} - \frac{1}{3} - \frac{0}{2} - \frac{0}{3} \right) \right]$$

$$= 2 \left[\frac{1}{6} - 0 \right]$$

$$a_0 = \frac{1}{3}$$

$$a_n = \frac{2}{L} \int_0^L f(x) \cos \left(\frac{n\pi x}{L} \right) dx$$

$$= \frac{2}{1} \int_0^1 f(x) \cdot \cos \left(\frac{n\pi x}{1} \right)$$

$$= \frac{2}{1} \int_0^1 (x-x^2) \cdot \cos(n\pi x)$$

$$= 2 \int_0^1 (x-x^2) \cdot \cos(n\pi x)$$

By Bernoulli's Rule,

$$= 2 \left[(x-x^2) \left(\frac{\sin(n\pi x)}{n\pi} \right) - (1-2x) \left(\frac{-\cos(n\pi x)}{n^2\pi^2} \right) + (0-2) \left(\frac{-\sin(n\pi x)}{n^3\pi^3} \right) \right]_0^1$$

$$a_n = 2 \left[\frac{(1-i)^2 \sin n\pi}{n\pi} + \frac{(1-2)(+\cos n\pi)}{n^2 \pi^2} + \frac{(0-2)(-\sin n\pi)}{n^3 \pi^3} \right] - \left[0 - (-1)(0) + 0 \right]$$

$$a_n = 2 \left[+(-1) \frac{(-1)^n}{n^2 \pi^2} + (0) - \left[\frac{(-1)}{n^2 \pi^2} \right] \right]$$

$$a_n = 2 \left[\frac{-(-1)^n - 1}{n^2 \pi^2} \right]$$

$$\therefore a_n = 2 \left(\frac{-(-1)^n - 1}{n^2 \pi^2} \right)$$

$$(x-x^2) = \frac{1}{6} + \sum_{n=1}^{\infty} \left[\left(2 \frac{(-1)^n}{n^2 \pi^2} - 1 \right) \cos(n\pi x) \right]$$

Ans:

$$(x-x^2) = \frac{1}{6} + \sum_{n=1}^{\infty} \left[\left(2 - \frac{(-1)^n - 1}{n^2 \pi^2} \right) \cdot \cos(n\pi x) \right]$$

Multiple Variable Calculus

Date: 19/10/24

Page: 1

Partial diff. equaⁿ. :

- The derivative of f^n of several variable w.r.t one of the independent variable keeping all remaining independent variable const. is called as Partial differenciaⁿ.

Let $Z = f(x, y)$ be a fn of two variable x & y where Z is a dependent variable and x, y are independent variable.

The derivative of Z w.r.t x keeping y const. is denoted by $\left(\frac{\partial Z}{\partial x}\right)_{y-\text{const.}}$

Ex - i) $Z = x^3 + y^3$

Partially diff. Z w.r.t x .

	keeping y constant	keeping x constant
$\rightarrow \therefore$	$\frac{\partial Z}{\partial x} = \frac{\partial (x^3 + y^3)}{\partial x}$	$\frac{\partial Z}{\partial y} = \frac{\partial (x^3 + y^3)}{\partial y}$

$$Z_x = 3x^2 + 0$$

$$\therefore \underline{Z_x = 3x^2}$$

$$Z_y = 0 + 3y^2$$

$$\therefore \underline{Z_y = 3y^2}$$

diff. w.r.t to y .

1. First order Partial Derivative :

let $u = f(x, y, z)$

$$u_x = \frac{\partial u}{\partial x}$$

$$u_y = \frac{\partial u}{\partial y}$$

$$u_z = \frac{\partial u}{\partial z}$$

2. Second order Partial Derivative :

$$u_{xx} = \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial x} \right) = \frac{\partial^2 u}{\partial x^2}$$

$$u_{yy} = \frac{\partial}{\partial y} \left(\frac{\partial u}{\partial y} \right) = \frac{\partial^2 u}{\partial y^2}$$

$$u_{xy} = \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial y} \right) = \frac{\partial^2 u}{\partial x \partial y}$$

$$u_{yx} = \frac{\partial}{\partial y} \left(\frac{\partial u}{\partial x} \right) = \frac{\partial^2 u}{\partial y \partial x}$$

$$u_{zz} = \frac{\partial}{\partial z} \left(\frac{\partial u}{\partial z} \right) = \frac{\partial^2 u}{\partial z^2}$$

$$u_{yz} = \frac{\partial}{\partial y} \left(\frac{\partial u}{\partial z} \right) = \frac{\partial^2 u}{\partial y \partial z}$$

$$U_{zy} = \frac{\partial}{\partial z} \left(\frac{\partial u}{\partial y} \right)$$

$$U_{zx} = \frac{\partial}{\partial z} \left(\frac{\partial u}{\partial x} \right)$$

$$U_{xz} = \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial z} \right)$$

3. Third order Partial Derivative :
Let $u = f(x, y)$

$$U_{xxx} = \frac{\partial}{\partial x} \left(\frac{\partial}{\partial x} \left(\frac{\partial u}{\partial x} \right) \right) = \frac{\partial^3 u}{\partial x^3}$$

$$U_{yyy} = \frac{\partial}{\partial y} \left(\frac{\partial}{\partial y} \left(\frac{\partial u}{\partial y} \right) \right) = \frac{\partial^3 u}{\partial y^3}$$

$$U_{xyy} = \frac{\partial}{\partial x} \left(\frac{\partial}{\partial y} \left(\frac{\partial u}{\partial y} \right) \right) =$$

$$U_{xxy} = \frac{\partial}{\partial x} \left(\frac{\partial}{\partial x} \left(\frac{\partial u}{\partial y} \right) \right) =$$

$$U_{xyx} = \frac{\partial}{\partial x} \left(\frac{\partial}{\partial y} \left(\frac{\partial u}{\partial x} \right) \right) =$$

$$U_{yyx} = \frac{\partial}{\partial y} \left(\frac{\partial}{\partial y} \left(\frac{\partial u}{\partial x} \right) \right) =$$

$$u_{yxy} = \frac{\partial}{\partial y} \left(\frac{\partial}{\partial x} \left(\frac{\partial u}{\partial y} \right) \right)$$

$$u_{yxx} = \frac{\partial}{\partial y} \left(\frac{\partial}{\partial x} \left(\frac{\partial u}{\partial x} \right) \right)$$

Rule of Partial Derivative :

1. Addition / Subtraction

Let $u = f(x, y)$ & $v = f(x, y)$

$$\frac{\partial}{\partial x} (u \pm v) = \frac{\partial u}{\partial x} \pm \frac{\partial v}{\partial x}$$

$$\frac{\partial}{\partial y} (u \pm v) = \frac{\partial u}{\partial y} \pm \frac{\partial v}{\partial y}$$

2. Product Rule

$$\frac{\partial}{\partial x} (u \cdot v) = u \frac{\partial v}{\partial x} + v \frac{\partial u}{\partial x}$$

$$\frac{\partial}{\partial y} (u \cdot v) = u \frac{\partial v}{\partial y} + v \frac{\partial u}{\partial y}$$

3. Quotient Rule

$$\frac{\partial}{\partial x} \left(\frac{u}{v} \right) = \frac{v \frac{\partial u}{\partial x} - u \frac{\partial v}{\partial x}}{v^2}$$

$$\frac{\partial}{\partial xy} \left(\frac{u}{v} \right) = \frac{v \cdot \frac{\partial u}{\partial y} - u \frac{\partial v}{\partial y}}{v^2}$$

4. Scalar Multiplication.

If k is any scalar

$$\frac{\partial (ku)}{\partial x} = k \frac{\partial u}{\partial x}$$

$$\frac{\partial (ku)}{\partial y} = k \frac{\partial u}{\partial y}$$

$$\frac{\partial (k)}{\partial x} = 0$$

$$\frac{\partial (k)}{\partial y} = 0$$

Some Standard formulae.

$$\log AB = \log A + \log B$$

$$\log \frac{A}{B} = \log A - \log B$$

$$\log m^n = n \log m$$

Ex-1 → If $Z = \log(x^2 + y^2)$, then find $\frac{\partial Z}{\partial x}$ & $\frac{\partial Z}{\partial y} = ?$

$$Z = \log(x^2 + y^2)$$

diff Z partially w.r.t x keeping y constant

$$\therefore \frac{\partial Z}{\partial x} = \frac{\partial (\log(x^2 + y^2))}{\partial x}$$

$$= \frac{1}{x^2 + y^2} (2x + 0)$$

$$\therefore \frac{\partial Z}{\partial x} = \frac{2x}{x^2 + y^2}$$

diff Z partially w.r.t y keeping x constant

$$\therefore \frac{\partial Z}{\partial y} = \frac{\partial (x^2 + y^2)}{\partial y}$$

$$\therefore \frac{\partial Z}{\partial y} = \frac{2y}{x^2 + y^2}$$

Ex-2 → If $Z = \tan^{-1}(y/x)$, find $\frac{\partial Z}{\partial x}$ & $\frac{\partial Z}{\partial y} = ?$

$$Z = \tan^{-1}(y/x)$$

diff Z w.r.t x keeping y constant.

$$\therefore \frac{\partial Z}{\partial x} = \frac{\partial (\tan^{-1} \frac{y}{x})}{\partial x}$$

$$\frac{\partial z}{\partial x} = \frac{1}{1 + (y/x)^2} \frac{\partial (y/x)}{\partial x}$$

$$= \frac{y}{1 + y^2} \cdot \frac{-1}{x^2}$$

$$\frac{\partial z}{\partial x} = \frac{y}{1 + y^2}$$

diff z w.r.t y keeping x constant.

$$\therefore \frac{\partial z}{\partial y} = \frac{\partial (\tan^{-1} y/x)}{\partial y} = \frac{1}{1 + y^2/x^2} \cdot \frac{1}{x}$$

$$= \frac{1}{1 + y^2/x^2} \cdot \frac{1}{x} = \frac{x}{x^2 + y^2}$$

$$\therefore \frac{\partial z}{\partial y} = \frac{x}{x^2 + y^2}$$

* Ex-3 If $z = x^y + y^x$ $\therefore$ find u_x & u_y

$$z = x^y + y^x$$

diff z w.r.t x keeping y constant.

$$\frac{\partial u}{\partial x} = \frac{\partial (x^y + y^x)}{\partial x}$$

$$= y(x^{y-1}) + y^x \log y$$

$$u_x = y(x^{y-1}) + y^x \log y$$

diff. u w.r.t y keeping x constant

$$\frac{\partial u}{\partial y} = \frac{\partial}{\partial y} (x^y + y^x)$$

$$u_y = x^y \cdot \log x + x y^{x-1}$$

Ex-4 IF $u = x^2 \cdot \tan^{-1}(y/x) - y^2 \cdot \tan^{-1}(x/y)$ find u_{xy}
 $\rightarrow$

$$u = x^2 \cdot \tan^{-1}(y/x) - y^2 \cdot \tan^{-1}(x/y)$$

$$u_{xy} = \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial y} \right)$$

diff. u w.r.t y keeping $x = \text{constant}$

$$\therefore \frac{\partial u}{\partial y} = x^2 \cdot \tan^{-1}\left(\frac{y}{x}\right) - \frac{\partial u}{\partial y} = y^2 \cdot \tan^{-1}\left(\frac{x}{y}\right)$$

by U.V. rule

$$= x^2 \cdot \frac{1}{1 + \frac{y^2}{x^2}} \cdot \frac{1}{x} - \left[y^2 \cdot \frac{1}{1 + \frac{x^2}{y^2}} \cdot \frac{\partial}{\partial y} \left(\frac{x}{y} \right) + \right.$$

$$\left. \tan^{-1}(x/y) \cdot 2y \right]$$

$$= \frac{x^2}{x^2 + y^2} \cdot \frac{1}{x} - \left[\frac{y^2}{y^2 + x^2} \cdot \frac{-x}{y^2} + \tan^{-1} \frac{x}{y} \cdot 2y \right]$$

$$= \frac{x^3}{x^2 + y^2} + \frac{xy^2}{x^2 + y^2} + \tan^{-1} \frac{x}{y} \cdot 2y$$

$$= x \left[\frac{x^2 + y^2}{x^2 + y^2} \right] + - \frac{2y \tan^{-1} x}{y}$$

$$\frac{\partial u}{\partial y} = x - 2y \tan^{-1} \frac{x}{y}$$

diff. w.r.t x keeping y constant.

$$u_{xy} = \frac{\partial u}{\partial x} = 1 - 2y \cdot \frac{1}{1 + \frac{x^2}{y^2}} \cdot \frac{\partial}{\partial x} \left(\frac{x}{y} \right)$$

$$= 1 - 2y \cdot \frac{y^2}{y^2 + x^2} \cdot \frac{1}{y}$$

$$= 1 - 2y \cdot \frac{y}{y^2 + x^2}$$

$$= \frac{1 - 2y^2}{y^2 + x^2}$$

$$= \frac{x^2 + y^2 - 2y^2}{y^2 + x^2} = \frac{x^2 - y^2}{x^2 + y^2}$$

$$u_{xy} = \frac{\partial u}{\partial x} = \frac{x^2 - y^2}{x^2 + y^2}$$

Ex-5 Find the value of n for which $z = t^n e^{-r^2/4t}$ satisfied partial diff. equⁿ

$$\frac{1}{r^2} \left[\left(\frac{\partial}{\partial r} \left(r^2 \frac{\partial z}{\partial r} \right) \right) \right] = \frac{\partial z}{\partial t}$$

→

$$\text{LHS} = \frac{1}{r^2} \left[\left(\frac{\partial}{\partial r} \left(r^2 \cdot \frac{\partial z}{\partial r} \right) \right) \right]$$

$$\frac{\partial z}{\partial r} = t^n e^{-r^2/4t} \quad \text{--- diff. } z \text{ w.r.t. } r$$

$$\frac{\partial z}{\partial r} = \frac{\partial (t^n)}{\partial r} \cdot e^{-r^2/4t}$$

$$= t^n \cdot e^{-r^2/4t} \cdot \frac{-2r}{2 \cdot 4t}$$

$$= -\frac{t^n}{2} \cdot e^{-r^2/4t} \cdot \frac{r}{t}$$

$$\frac{\partial z}{\partial r} = -\frac{t^{n-1}}{2} \cdot r \cdot e^{-r^2/4t}$$

multiply by r^2 , to $2r$ on both sides

$$\frac{r^2 \cdot \partial z}{\partial r} = -\frac{t^{n-1}}{2} \cdot r^3 \cdot e^{-r^2/4t}$$

part. diff w.r.t to r

$$\frac{\partial}{\partial r} \left(r^2 \cdot \frac{\partial z}{\partial r} \right) = \frac{\partial}{\partial r} \left(-\frac{t^{n-1}}{2} \cdot r^3 \cdot e^{-r^2/4t} \right)$$

$$= -\frac{t^{n-1}}{2} \cdot \frac{\partial}{\partial r} \left(r^3 \cdot e^{-r^2/4t} \right)$$

$$= -\frac{t^{n-1}}{2} \left[r^3 \cdot e^{-r^2/4t} \cdot \left(-\frac{2r}{4t} \right) + e^{-r^2/4t} \cdot 3r^2 \right]$$

$$= -\frac{t^{n-1}}{2} \cdot e^{-r^2/4t} \cdot r^2 \left[\frac{-r^2}{2t} + 3 \right]$$

Multiplying by $1/r^2$ on both sides,

$$= \frac{1}{r^2} \left[-\frac{t^{n-1}}{2} \cdot e^{-r^2/4t} \cdot r^2 \left(\frac{-r^2}{2t} + 3 \right) \right]$$

$$\text{LHS} = -\frac{t^{n-1}}{2} e^{-r^2/4t} \left(\frac{-r^2}{2t} + 3 \right) \quad \text{--- (1)}$$

$$\text{RHS} = \frac{\partial z}{\partial t}$$

$$= \frac{\partial}{\partial t} \left(t^n \cdot e^{-r^2/4t} \right) \quad \text{--- diff. } z \text{ w.r.t } t \text{ keeping } r \text{ constant}$$

$$= t^n \cdot e^{-r^2/4t} \cdot \frac{-r^2}{4t^2} + e^{-r^2/4t} \cdot n t^{n-1}$$

$$= t^n \cdot e^{-r^2/4t} \cdot \frac{\partial^2}{\partial t^2} \left(\frac{1}{t^2} \right) + n t^{n-1} \cdot e^{-r^2/4t}$$

$$= e^{-r^2/4t} \cdot t^n \cdot t^{-2} \cdot \frac{r^2}{4} + n t^{n-1} \cdot e^{-r^2/4t}$$

$$= e^{-r^2/4t} \cdot t^{n-1} \left[\frac{t^{n-1} \cdot r^2}{4t} + n \right]$$

$$\text{RHS} = t^{n-1} \cdot e^{-r^2/4t} \left(\frac{r^2}{4t} + n \right) \quad \text{--- (2)}$$

$$\text{LHS} = \text{RHS}$$

$$-\frac{t^{n-1}}{2} e^{-r^2/4t} \left(\frac{-r^2}{2t} + 3 \right) = t^{n-1} e^{-r^2/4t} \left(\frac{r^2}{4t} + n \right)$$

$$\left(\frac{-1}{2} \right) \left(t^n t^{n-1} \right) e^{-r^2/4t} \left(\frac{-r^2}{2t} + 3 \right) = t^{n-1} e^{-r^2/4t} \left(\frac{r^2}{4t} + n \right)$$

$$+\frac{r^2}{4t} - \frac{3}{2} = \frac{r^2}{4t} + n$$

$$\therefore \boxed{n = -3/2}$$

Ex-6 P.t. a point of a surface $x^x \cdot y^y \cdot z^z = c$ where $x = y = z$, then

$$\frac{\partial^2 z}{\partial x \partial y} = -(x \log x)^{-1}$$

→

$$x^x y^y z^z = c$$

taking log on both sides

$$\log(x^x y^y z^z) = \log c$$

$$\log x^x + \log y^y + \log z^z = \log c$$

$$x \log x + y \log y + z \log z = \log c \quad \text{--- (1)}$$

diff w.r.t y keeping x constant

$$\frac{\partial}{\partial y} x \log x + \frac{\partial}{\partial y} y \log y + \frac{\partial}{\partial y} z \log z = \frac{\partial}{\partial y} \log c$$

$x = \text{constant}$

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$$0 + y \cdot \frac{1}{y} + \log y + z \cdot \frac{1}{z} \frac{\partial z}{\partial y} + \log z \frac{\partial z}{\partial y} = 0$$

$$(1 + \log y) + \frac{\partial z}{\partial y} (1 + \log z) = 0$$

$$= \frac{\partial z}{\partial y} (1 + \log z) = - (1 + \log y)$$

$$\therefore \frac{\partial z}{\partial y} = - \frac{(1 + \log y)}{(1 + \log z)}$$

we diff w.r.t x keeping y - constant.

$$\frac{\partial}{\partial x} \left(\frac{\partial z}{\partial y} \right) = \frac{\partial}{\partial x} \left(\frac{-(1 + \log y)}{(1 + \log z)} \right)$$

$$\frac{\partial^2 z}{\partial xy} = - (1 + \log y) \frac{\partial}{\partial x} \left(\frac{1}{1 + \log x} \right)$$

$$\frac{\partial^2 z}{\partial xy} = - (1 + \log y) \left(- \frac{1}{(1 + \log x)^2} \frac{\partial}{\partial x} (1 + \log x) \right)$$

$$\frac{\partial^2 z}{\partial xy} = \frac{(1 + \log y)}{(1 + \log x)^2} \cdot \frac{1}{z} \cdot \frac{\partial z}{\partial x} \quad \text{--- (2)}$$

diff. equ. (1) w.r.t to x

$$\frac{\partial}{\partial x} (x \log x) + \frac{\partial}{\partial x} (y \cdot \log y) + \frac{\partial}{\partial x} (z \cdot \log z) = \frac{\partial}{\partial x} (\log c)$$

$$x \cdot \frac{1}{x} + \log x + 2 \cdot \frac{1}{z} \frac{\partial z}{\partial x} + \log z \frac{\partial z}{\partial x} = 0$$

$$(1 + \log x) + \frac{\partial z}{\partial x} (1 + \log z) = 0$$

$$\frac{\partial z}{\partial x} (1 + \log z) = -(1 + \log x)$$

$$\frac{\partial z}{\partial x} = - \left(\frac{1 + \log x}{1 + \log z} \right)$$

put value of $\partial z / \partial x$ in eqn. (2)

$$= \frac{(1 + \log y)}{(1 + \log z)^2} \cdot \frac{1}{z} - \frac{(1 + \log x)}{(1 + \log z)}$$

$$= \frac{-(1 + \log x) \cdot (1 + \log y)}{z (1 + \log z)^3}$$

$$= \frac{-(1 + \log x) (1 + \log x)}{x (1 + \log x)^3} \quad \because x = y = z$$

$$= \frac{-1}{x (1 + \log x)} = \frac{-1}{x (\log e + \log x)}$$

$$= \frac{-1}{x (\log ex)} \quad \because \log a + \log b = \log ab$$

$$= -(x (\log ex))^{-1}$$

Hence proved.

constant

add./subt. - 0

mult./divis - as it is baher

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Example on Verification :

Ex-1. $\rightarrow$ If $u = \log(x^2 + y^2)$ then verify $u_{xy} = u_{yx}$

$$u_{xy} = \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial y} \right)$$

diff u w.r.t y keeping x constant.

$$\frac{\partial u}{\partial y} = \frac{\partial}{\partial y} (\log(x^2 + y^2))$$

$$\frac{\partial u}{\partial y} = \frac{1}{x^2 + y^2} \cdot 2y = \frac{2y}{x^2 + y^2}$$

$$\therefore \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial y} \right) = \frac{\partial}{\partial x} \left(\frac{2y}{x^2 + y^2} \right)$$

--- diff. w.r.t x
keeping y const.

$$= 2y \left(\frac{-1}{(x^2 + y^2)^2} \right) \cdot 2x$$

$$\therefore \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial y} \right) = \frac{-2yx}{(x^2 + y^2)^2} \quad \text{--- (1)}$$

$$\therefore u_{yx} = \frac{\partial}{\partial y} \left(\frac{\partial u}{\partial x} \right)$$

$$\therefore \frac{\partial u}{\partial x} = \frac{\partial}{\partial x} (\log(x^2 + y^2))$$

--- diff. w.r.t x
keeping y const.

$$\frac{\partial u}{\partial x} = \frac{1}{x^2 + y^2} \cdot 2x + 0 = \frac{2x}{x^2 + y^2}$$

$$\frac{\partial}{\partial y} \left(\frac{\partial u}{\partial x} \right) = \frac{\partial}{\partial y} \left(\frac{2x}{x^2+y^2} \right)$$

$$= 2x \cdot \frac{\partial}{\partial y} \left(\frac{1}{x^2+y^2} \right)$$

$$= 2x \cdot \frac{-1}{(x^2+y^2)^2} \cdot 2y$$

$$= \frac{-4xy}{(x^2+y^2)^2} \quad (2)$$

from (1) and (2)

$u_{xy} = v_{yx}$ Hence Verified.

Ex-2 If $u = \tan(y+ax) - (y-ax)^{3/2}$ verify $u_{xy} = u_{yx}$
 $\rightarrow u = \tan(y+ax) - (y-ax)^{3/2}$

$$u_{xy} = \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial y} \right) = \frac{\partial}{\partial x} \left(\frac{\partial}{\partial y} [\tan(y+ax) - (y-ax)^{3/2}] \right)$$

$$\therefore \frac{\partial u}{\partial y} = \frac{\partial}{\partial y} \tan(y+ax) - \frac{\partial}{\partial y} (y-ax)^{3/2}$$

$$= \sec^2(y+ax) - \frac{3}{2}(1-a)^{3/2-1}$$

$$= \sec^2(y+ax) - \frac{3}{2} 1^{3/2}$$

$$= \sec^2(y+ax) - \frac{3}{2} (y-ax)^{1/2}$$

$$\frac{\partial u}{\partial y} = \sec^2(y+ax) - \frac{3}{2} (y-ax)^{1/2}$$

$$\frac{\partial}{\partial x} \left(\frac{\partial u}{\partial y} \right) = \frac{\partial}{\partial x} \left(\sec^2(y+ax) - \frac{3}{2} (y-ax)^{1/2} \right)$$

$$= 2 \sec(y+ax) \cdot \sec(y+ax) \cdot \tan(y+ax) \cdot a - \frac{3}{2} \cdot \frac{1}{2} (y-ax)^{-1/2} \cdot (-a)$$

$$U_{xy} = 2 \sec^2(y+ax) \cdot \tan(y+ax) \cdot a + \frac{3a}{4} (y-ax)^{-1/2}$$

— (1)

$$\therefore U_{yx} = \frac{\partial}{\partial y} \left(\frac{\partial u}{\partial x} \right)$$

$$\therefore \frac{\partial u}{\partial x} = \frac{\partial}{\partial x} \left(\tan(y+ax) - (y-ax)^{3/2} \right)$$

$$\frac{\partial u}{\partial x} = \sec^2(y+ax) \cdot a - \frac{3}{2} (y-ax)^{1/2} \cdot (-a)$$

$$\frac{\partial u}{\partial x} = a \sec^2(y+ax) + \frac{3a}{2} (y-ax)^{1/2}$$

$$\frac{\partial}{\partial y} \left(\frac{\partial u}{\partial x} \right) = a \cdot 2 \sec(y+ax) \cdot \sec(y+ax) \cdot \tan(y+ax) + \left(\frac{3a}{2} \cdot \frac{1}{2} (y-ax)^{-1/2} \right)$$

$$= 2a \sec(y+ax) \cdot \sec(y+ax) \cdot \tan(y+ax) + \frac{3a}{4} (y-ax)^{-1/2}$$

$$U_{yx} = 2a \sec^2(y+ax) \cdot \tan(y+ax) + \frac{3a}{4} (y-ax)^{-1/2}$$

Hence verified

Ex-3 $U = \log(\tan x + \tan y + \tan z)$ s.t.
 $\sin 2x \cdot u_x + \sin 2y \cdot u_y + \sin 2z \cdot u_z = 2$

→ $u_x = \frac{\partial U}{\partial x} = \frac{1}{\tan x + \tan y + \tan z} \cdot \sec^2 x$

- diff. U w.r.t. x keeping y & z const.

$$= \frac{1}{\tan x + \tan y + \tan z} \cdot \sec^2 x$$

$u_y = \frac{\partial U}{\partial y} = \frac{1}{\tan x + \tan y + \tan z} \cdot \sec^2 y$ keeping x & z const.

$u_z = \frac{\partial U}{\partial z} = \frac{1}{\tan x + \tan y + \tan z} \cdot \sec^2 z$ keeping x & y const.

From given equaⁿ.

LHS = $\sin 2x \cdot \frac{\sec^2 x}{\tan x + \tan y + \tan z} + \sin 2y \cdot \frac{\sec^2 y}{\tan x + \tan y + \tan z}$

+ $\sin 2z \cdot \frac{\sec^2 z}{\tan x + \tan y + \tan z}$

= $2 \sin x \cdot \cos x \cdot \frac{1}{\cos^2 x} + 2 \sin y \cdot \cos y \cdot \frac{1}{\cos^2 y} +$

$\frac{2 \sin x \cdot \cos x}{\tan x + \tan y + \tan z} + \frac{2 \sin y \cdot \cos y}{\tan x + \tan y + \tan z} +$

$\frac{2 \sin z \cdot \cos z}{\tan x + \tan y + \tan z} \cdot \frac{1}{\cos^2 z}$

$\frac{2 \sin x \cdot \cos x}{\tan x + \tan y + \tan z} + \frac{2 \sin y \cdot \cos y}{\tan x + \tan y + \tan z} + \frac{2 \sin z \cdot \cos z}{\tan x + \tan y + \tan z}$

$$= \frac{2 \tan x + 2 \tan y + 2 \tan z}{\tan x + \tan y + \tan z}$$

$$= \frac{2 (\tan x + \tan y + \tan z)}{(\tan x + \tan y + \tan z)}$$

$$\text{RHS} = 2$$

$$\therefore \text{LHS} = \text{RHS} \quad \text{Hence shown.}$$

Ex-4 If $U = \log(x^3 + y^3 - y^2x - x^2y)$. pt.

$$\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} \right)^2 U = \frac{-4}{(x+y)^2}$$

$$\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} \right) \left(\frac{\partial U}{\partial x} + \frac{\partial U}{\partial y} \right)$$

$$\therefore \frac{\partial}{\partial x} (\log(x^3 + y^3 - y^2x - x^2y)) \quad \text{--- diff. } U \text{ w.r.t } x \text{ partially}$$

$$\frac{\partial U}{\partial x} = \frac{3x^2 + 0 - y^2 - 2xy}{x^3 + y^3 - y^2x - x^2y}$$

$$\frac{\partial U}{\partial x} = \frac{3x^2 - y^2 - 2xy}{x^3 + y^3 - y^2x - x^2y} \quad \text{--- (1)}$$

$$\frac{\partial U}{\partial y} = \frac{3y^2 - 2yx - x^2}{x^3 + y^3 - y^2x - x^2y} \quad \text{--- (2)}$$

Now, add ① & ②.

$$\frac{3x^2 - y^2 - 2xy}{x^3 + y^3 - y^2x - x^2y} + \frac{3y^2 - 2yx - x^2}{x^3 + y^3 - y^2x - x^2y}$$

$$= \frac{3x^2 + 3y^2 - y^2 - 2xy - 2yx - x^2}{(x^3 + y^3 - y^2x - x^2y)}$$

$$= \frac{2x^2 + 2y^2 - 4xy}{x^3 + y^3 - y^2x - x^2y}$$

$$= \frac{2(x^2 + y^2 - 2xy)}{x^3 - x^2y + y^3 - y^2x} = \frac{2(x-y)^2}{x^2(x-y) + y^2(y-x)}$$

$$= \frac{2(x-y)^2}{x^2(x-y) - y^2(x-y)}$$

$$\begin{aligned} \text{common} & \quad (x-y) = \frac{2(x-y)^2}{(x-y)^2(x^2 - y^2)} = \frac{2(x-y)^2}{(x+y)(x-y)(x-y)} \\ \text{nikalo} & \end{aligned}$$

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = \frac{2}{(x+y)}$$

$$\therefore \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} \right) \left(\frac{2}{x+y} \right)$$

$$= \frac{\partial}{\partial x} \left(\frac{2}{x+y} \right) + \frac{\partial}{\partial y} \left(\frac{2}{x+y} \right)$$

$$= 2 \cdot \frac{-1}{(x+y)^2} + 2 \left(\frac{-1}{(x+y)^2} \right)$$

$$= \frac{-2}{(x+y)^2} - \frac{2}{(x+y)^2}$$

$$= \frac{-4}{(x+y)^2}$$

Hence proved.

Ex-5 If $U = \log \sqrt{x^2 + y^2 + z^2}$ then p.t.

$$(x^2 + y^2 + z^2) \left(\frac{\partial^2 U}{\partial x^2} + \frac{\partial^2 U}{\partial y^2} + \frac{\partial^2 U}{\partial z^2} \right) = 1$$

→

Composite Function :Derivative of composite f^n .

1) if $z = f(u)$

$u = f(x, y)$

i.e. z is a f^n of u , & u is a f^n of x, y .

then,

 z is called as composite f^n of x, y .Note:

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} = \frac{dz}{du} \cdot \frac{\partial u}{\partial x}$$

$$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial y} = \frac{dz}{du} \cdot \frac{\partial u}{\partial y}$$

2) if $z = u \cdot v$

$(u, v) = (x, y)$

i.e. z is f^n of u, v . u is f^n of x, y .

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x}$$

$$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial y}$$

3) if $z = u, v$, $u \cdot v = t$

$$\therefore \frac{dz}{dt} = \frac{\partial z}{\partial u} \cdot \frac{du}{dt} + \frac{\partial z}{\partial v} \cdot \frac{dv}{dt}$$

If w.r.t. f is not present in f^n , then the derivative is zero.

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$$x = u+v, \quad y = u \cdot v$$

Ex-1 if z is $f(x, y)$ where

$$\text{find } \frac{\partial z}{\partial u} \text{ \& } \frac{\partial z}{\partial v}$$

— (1)

$$\rightarrow \frac{\partial z}{\partial u} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial u}$$

$$z \rightarrow x \cdot y \rightarrow u \cdot v$$

$$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial v}$$

— (2)

from equ (1)

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x} \cdot 1 + \frac{\partial z}{\partial y} (v)$$

Ans: $\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x} + v \cdot \frac{\partial z}{\partial y}$

from equ (2)

$$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x} \cdot 1 + \frac{\partial z}{\partial y} (u)$$

Ans: $\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x} + u \cdot \frac{\partial z}{\partial y}$

Ex-2 If $\phi = f(x, y, z)$ where $x = u+v+w$,

$$y = uv + vw + uw, \quad z = uvw$$

$$\frac{\partial \phi}{\partial u}, \quad \frac{\partial \phi}{\partial v}, \quad \frac{\partial \phi}{\partial w}$$

$$\rightarrow \frac{\partial \phi}{\partial u} = \frac{\partial \phi}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial \phi}{\partial y} \cdot \frac{\partial y}{\partial u} + \frac{\partial \phi}{\partial z} \cdot \frac{\partial z}{\partial u} \quad \text{— (3)}$$

$$\phi \rightarrow x, y, z \rightarrow u, v, w$$

$$\frac{\partial \phi}{\partial v} = \frac{\partial \phi}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial \phi}{\partial y} \cdot \frac{\partial y}{\partial v} + \frac{\partial \phi}{\partial z} \cdot \frac{\partial z}{\partial v} \quad \text{--- (2)}$$

$$\frac{\partial \phi}{\partial w} = \frac{\partial \phi}{\partial x} \cdot \frac{\partial x}{\partial w} + \frac{\partial \phi}{\partial y} \cdot \frac{\partial y}{\partial w} + \frac{\partial \phi}{\partial z} \cdot \frac{\partial z}{\partial w} \quad \text{--- (3)}$$

$\therefore$ from equ. (1)

$$\frac{\partial \phi}{\partial u} = \frac{\partial \phi}{\partial x} \cdot 1 + \frac{\partial \phi}{\partial y} \cdot v + v w + w + \frac{\partial \phi}{\partial z} \cdot v w$$

Ans: $\therefore \frac{\partial \phi}{\partial u} = \frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} (v + w + vw) + v w \cdot \frac{\partial \phi}{\partial z}$

$\therefore$ from equ. (2)

$$\frac{\partial \phi}{\partial v} = \frac{\partial \phi}{\partial x} \cdot 1 + \frac{\partial \phi}{\partial y} (u + w + uw) + \frac{\partial \phi}{\partial z} \cdot (u w)$$

Ans: $\therefore \frac{\partial \phi}{\partial v} = \frac{\partial \phi}{\partial x} + (u + w + uw) \cdot \frac{\partial \phi}{\partial y} + (u w) \cdot \frac{\partial \phi}{\partial z}$

$\therefore$ from equ. (3)

$$\frac{\partial \phi}{\partial w} = \frac{\partial \phi}{\partial x} \cdot \frac{\partial x}{\partial w} + \frac{\partial \phi}{\partial y} \cdot \frac{\partial y}{\partial w} + \frac{\partial \phi}{\partial z} \cdot \frac{\partial z}{\partial w}$$

$$\frac{\partial \phi}{\partial w} = \frac{\partial \phi}{\partial x} \cdot 1 + \frac{\partial \phi}{\partial y} (u + v + uv) + \frac{\partial \phi}{\partial z} \cdot (u v)$$

Ans: $\therefore \frac{\partial \phi}{\partial w} = \frac{\partial \phi}{\partial x} + (u + v + uv) \cdot \frac{\partial \phi}{\partial y} + (u v) \cdot \frac{\partial \phi}{\partial z}$

Ex-3 IF $u = f(r)$ & $r = \sqrt{x^2 + y^2}$, then p.t.

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f''(r) + \frac{1}{r} f'(r)$$

→

$$u \rightarrow r \rightarrow x, y \quad \because x^2 + y^2 = r^2 \quad +$$

$$r = \sqrt{x^2 + y^2}$$

$$\frac{\partial r}{\partial x} = \frac{\partial (\sqrt{x^2 + y^2})}{\partial x} \quad \text{--- diff. w.r.t } x \text{ keeping } y \text{ constant}$$

$$\frac{\partial r}{\partial x} = \frac{1}{2\sqrt{x^2 + y^2}} (2x) = \frac{x}{\sqrt{x^2 + y^2}}$$

$$\frac{\partial r}{\partial x} = \frac{x}{\sqrt{x^2 + y^2}} = \frac{x}{r} \quad \therefore \frac{\partial r}{\partial x} = \frac{x}{r} \quad \text{--- (1)}$$

$$\text{Similarly, } \frac{\partial r}{\partial y} = \frac{y}{r} \quad \text{--- (2)}$$

$$u = f(r)$$

--- diff. u w.r.t. x , partially,

$$\frac{\partial u}{\partial x} = \frac{\partial (f(r))}{\partial x}$$

$$= \frac{\partial f'(r)}{\partial x} \cdot \frac{\partial r}{\partial x}$$

$$= f'(r) \cdot \frac{x}{r}$$

$$\frac{\partial u}{\partial x} = f'(r) \cdot x \cdot r^{-1}$$

diff. partially again w.r.t. x .

$$\therefore \frac{\partial^2 u}{\partial x^2} = \frac{\partial (f'(r) \cdot x \cdot r^{-1})}{\partial x^2} \quad \text{U.V.W.}$$

$$= x \cdot r^{-1} \cdot f''(r) \cdot \frac{\partial r}{\partial x} + f'(r) \cdot r^{-1} \cdot 1 +$$

$$f'(r) \cdot x \cdot (-r^{-2}) \cdot \frac{\partial r}{\partial x}$$

$$= x \cdot r^{-1} \cdot f''(r) \cdot x + f'(r) \cdot r^{-1} + f'(r) \cdot x \cdot$$

$$- r^{-2} \cdot x$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{x^2 f''(r)}{r^2} + \frac{f'(r)}{r} - \frac{f'(r) \cdot x^2}{r^3} \quad \text{--- (1)}$$

|| y,

$$\frac{\partial^2 u}{\partial y^2} = \frac{y^2 f''(r)}{r^2} + \frac{f'(r)}{r} - \frac{f'(r) \cdot y^2}{r^3} \quad \text{--- (2)}$$

$$\therefore \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} =$$

$$= \frac{x^2 f''(r)}{r^2} + \frac{f'(r)}{r} - \frac{f'(r) \cdot x^2}{r^3} +$$

$$\frac{y^2 f''(r)}{r^2} + \frac{f'(r)}{r} - \frac{f'(r) \cdot y^2}{r^3}$$

$$= \frac{f''(r) \cdot (x^2 + y^2)}{r^2} + \frac{2 \cdot f'(r)}{r} - \frac{f'(r) \cdot (x^2 + y^2)}{r^3}$$

$$= \frac{f''(\alpha) \alpha^2}{\alpha^2} + \frac{2f'(\alpha)}{\alpha} - \frac{f'(\alpha)}{\alpha^3} \cdot \alpha^2$$

$$= f''(\alpha) + \frac{2f'(\alpha)}{\alpha} - \frac{f'(\alpha)}{\alpha}$$

$$\text{RHS} = f''(\alpha) + \frac{2f'(\alpha)}{\alpha}$$

$$\therefore \text{LHS} = \text{RHS}$$

Hence proved.

Ex-4 $z = f(x, y)$ & $x = r \cos \theta$, $y = r \sin \theta$ p.t.

$$\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 = \left(\frac{\partial z}{\partial r}\right)^2 + \frac{1}{r^2} \left(\frac{\partial z}{\partial \theta}\right)^2$$

→

$$z \rightarrow x, y \rightarrow r, \theta$$

$$x = r \cos \theta, \quad y = r \sin \theta$$

$$\frac{\partial x}{\partial r} = \frac{\partial (r \cos \theta)}{\partial r}$$

— diff. x w.r.t r ,
keeping θ constant.

$$\frac{\partial x}{\partial r} = \cos \theta \quad \text{--- (1)}$$

$$\text{Ily } \frac{\partial y}{\partial r} = \sin \theta \quad \text{--- (2)}$$

$$z = f(x, y)$$

diff z w.r.t x

Keeping y const.

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial r} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial x}$$

$$\frac{\partial z}{\partial r} = \frac{\partial z}{\partial x} \cdot \cos \theta + \frac{\partial z}{\partial y} \cdot \sin \theta \quad \text{--- (3)}$$

$$\begin{aligned} \frac{\partial z}{\partial \theta} &= \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial \theta} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial \theta} \\ &= \frac{\partial z}{\partial x} \cdot (-r \sin \theta) + \frac{\partial z}{\partial y} \cdot r \cos \theta \quad \text{--- (4)} \end{aligned}$$

$$\frac{\partial z}{\partial \theta} = r \left[\frac{\partial z}{\partial x} \cdot (-\sin \theta) + \frac{\partial z}{\partial y} \cdot \cos \theta \right]$$

$$\therefore \frac{1}{r} \cdot \frac{\partial z}{\partial \theta} = \left[\frac{\partial z}{\partial x} \cdot (-\sin \theta) + \frac{\partial z}{\partial y} \cdot \cos \theta \right] \quad \text{--- (4)}$$

Squaring and adding eqn. (3) & (4)

$$\left(\frac{\partial z}{\partial r} \right)^2 + \frac{1}{r^2} \left(\frac{\partial z}{\partial \theta} \right)^2 = \left(\frac{\partial z}{\partial x} \right)^2 \cos^2 \theta + \left(\frac{\partial z}{\partial y} \right)^2 \sin^2 \theta$$

$$+ \left(\frac{\partial z}{\partial x} \right)^2 (-\sin \theta)^2 + \left(\frac{\partial z}{\partial y} \right)^2 \cos^2 \theta$$

$$= \left(\frac{\partial z}{\partial x} \right)^2 (\cos^2 \theta + \sin^2 \theta) + \left(\frac{\partial z}{\partial y} \right)^2 (\sin^2 \theta + \cos^2 \theta)$$

$$\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 = \left(\frac{\partial z}{\partial \theta}\right)^2 + \frac{1}{r} \left(\frac{\partial z}{\partial \theta}\right)^2$$

$\therefore \text{LHS} = \text{RHS}$

Hence proved.

Ex- if $z = f(x, y)$, $x = e^u + e^{-v}$, $y = e^{-u} + e^v$ p.1

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial u}$$

→

$$z \rightarrow x, y \rightarrow u, v$$

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial u}$$

$$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial v}$$

$$\frac{\partial z}{\partial x} = \frac{\partial (f(x, y))}{\partial x}$$

=

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x} \cdot \frac{\partial (e^u + e^{-v})}{\partial u} + \frac{\partial z}{\partial y} \cdot \frac{\partial (e^{-u} + e^v)}{\partial u}$$

$$= \frac{\partial z}{\partial x} \cdot (e^u + 0) + \frac{\partial z}{\partial y} \cdot (-e^{-u})$$

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x} \cdot e^u + \frac{\partial z}{\partial y} \cdot (-e^{-u}) \quad \text{--- (1)}$$

$$\begin{aligned}\frac{\partial z}{\partial v} &= \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial v} \\ &= \frac{\partial z}{\partial x} \cdot \frac{\partial (e^u + e^{-v})}{\partial v} + \frac{\partial z}{\partial y} \cdot \frac{\partial (e^{-u} - e^v)}{\partial v}\end{aligned}$$

$$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x} \cdot (-e^{-v}) + \frac{\partial z}{\partial y} \cdot (-e^v) \quad \text{--- (2)}$$

Subtract equ. (2) from (1)

$$\begin{aligned}\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} &= \left[\frac{\partial z}{\partial x} \cdot e^u - \frac{\partial z}{\partial y} \cdot e^{-u} \right] - \left[\frac{\partial z}{\partial x} \cdot (-e^{-v}) + \frac{\partial z}{\partial y} \cdot (-e^v) \right] \\ &= e^u \frac{\partial z}{\partial x} - \frac{\partial z}{\partial y} e^{-u} + \frac{\partial z}{\partial x} e^{-v} + e^v \frac{\partial z}{\partial y} \\ &= \frac{\partial z}{\partial x} (e^u + e^{-v}) + \frac{\partial z}{\partial y} (e^v - e^{-u}) \\ &= \frac{\partial z}{\partial x} (e^u + e^{-v}) - \frac{\partial z}{\partial y} (e^{-u} - e^v)\end{aligned}$$

$$\text{RHS} = x \cdot \frac{\partial z}{\partial x} - y \cdot \frac{\partial z}{\partial y}$$

$$\therefore \text{LHS} = \text{RHS}$$

Hence proved.

Unit
ImpEx-1 If $u = f(x-y, y-z, z-x)$ Find,

$$\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial u}{\partial z}$$

→ Let $l = x-y$, $m = y-z$, $n = z-x$

$u \rightarrow l, m, n$ i.e. $u = f(l, m, n)$

$u \rightarrow l, m, n \rightarrow x, y, z$

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial l} \cdot \frac{\partial l}{\partial x} + \frac{\partial u}{\partial m} \cdot \frac{\partial m}{\partial x} + \frac{\partial u}{\partial n} \cdot \frac{\partial n}{\partial x}$$

$$= \frac{\partial u}{\partial l} \cdot \frac{\partial (x-y)}{\partial x} + \frac{\partial u}{\partial m} \cdot \frac{\partial (y-z)}{\partial x} + \frac{\partial u}{\partial n} \cdot \frac{\partial (z-x)}{\partial x}$$

$$= \frac{\partial u}{\partial l} \cdot 1 + \frac{\partial u}{\partial m} \cdot (0) + \frac{\partial u}{\partial n} \cdot (-1)$$

Ans: $\frac{\partial u}{\partial x} = \frac{\partial u}{\partial l} - \frac{\partial u}{\partial n}$ — (1)

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial l} \cdot \frac{\partial l}{\partial y} + \frac{\partial u}{\partial m} \cdot \frac{\partial m}{\partial y} + \frac{\partial u}{\partial n} \cdot \frac{\partial n}{\partial y}$$

$$= \frac{\partial u}{\partial l} \cdot (-1) + \frac{\partial u}{\partial m} \cdot (1) + \frac{\partial u}{\partial n} \cdot (0)$$

Ans: $\frac{\partial u}{\partial y} = -\frac{\partial u}{\partial l} + \frac{\partial u}{\partial m}$ — (2)

$$\frac{\partial u}{\partial z} = \frac{\partial u}{\partial l} \cdot \frac{\partial l}{\partial z} + \frac{\partial u}{\partial m} \cdot \frac{\partial m}{\partial z} + \frac{\partial u}{\partial n} \cdot \frac{\partial n}{\partial z}$$

$$= \frac{\partial u}{\partial l} \cdot 0 + \frac{\partial u}{\partial m} (-1) + \frac{\partial u}{\partial n} (1)$$

Ans: $\frac{\partial u}{\partial z} = -\frac{\partial u}{\partial m} + \frac{\partial u}{\partial n}$ — (3)

Ex-2 If $U = (x/y, y/z, z/x)$ then find, $\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial u}{\partial z}$

→

Let $x/y = a, y/z = b, z/x = c$

$U \rightarrow a, b, c \rightarrow x, y, z$

$U \rightarrow a, b, c \rightarrow x, y, z$

$$\frac{\partial u}{\partial a}$$

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial a} \cdot \frac{\partial a}{\partial x} + \frac{\partial u}{\partial b} \cdot \frac{\partial b}{\partial x} + \frac{\partial u}{\partial c} \cdot \frac{\partial c}{\partial x}$$

$$\therefore \frac{\partial u}{\partial x} = \frac{\partial u}{\partial a} \cdot \frac{1}{y} + \frac{\partial u}{\partial b} \cdot 0 + \frac{\partial u}{\partial c} \cdot \frac{z}{x^2} \cdot (-1)$$

$$\therefore \frac{\partial u}{\partial x} = \frac{1}{y} \cdot \frac{\partial u}{\partial a} + \left(-\frac{z}{x^2} \cdot \frac{\partial u}{\partial c} \right)$$

Ans: $\therefore \frac{\partial u}{\partial x} = \frac{1}{y} \cdot \frac{\partial u}{\partial a} - \frac{z}{x^2} \cdot \frac{\partial u}{\partial c}$ — (4)

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial a} \cdot \frac{\partial a}{\partial y} + \frac{\partial u}{\partial b} \cdot \frac{\partial b}{\partial y} + \frac{\partial u}{\partial c} \cdot \frac{\partial c}{\partial y}$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial a} \cdot x \cdot \frac{-1}{y^2} + \frac{\partial u}{\partial b} \cdot \frac{1}{z} + \frac{\partial u}{\partial c} (0)$$

Ans: $\frac{\partial u}{\partial y} = \frac{-x}{y^2} \cdot \frac{\partial u}{\partial a} + \frac{1}{z} \cdot \frac{\partial u}{\partial b} + \frac{\partial u}{\partial c} (0)$ — (2)

$$\frac{\partial u}{\partial z} = \frac{\partial u}{\partial a} \cdot \frac{\partial a}{\partial z} + \frac{\partial u}{\partial b} \cdot \frac{\partial b}{\partial z} + \frac{\partial u}{\partial c} \cdot \frac{\partial c}{\partial z}$$

$$= \frac{\partial u}{\partial a} (0) + \frac{\partial u}{\partial b} \cdot y \cdot \frac{-1}{z^2} + \frac{\partial u}{\partial c} \cdot \frac{1}{x}$$

Ans: $\frac{\partial u}{\partial z} = \frac{\partial u}{\partial b} \cdot \frac{-y}{z^2} + \frac{\partial u}{\partial c} \cdot \frac{1}{x}$ — (3)

Ex-3 if $v = f(e^{x-y}, e^{y-x}, e^{z-x})$ then s.t.

$$\frac{\partial v}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial v}{\partial z} = 0$$

Let $e^{x-y} = a$, $e^{y-x} = b$, $e^{z-x} = c$

$$y \rightarrow a, b, c \rightarrow x, y, z$$

$$\therefore \frac{\partial v}{\partial x} = \frac{\partial v}{\partial a} \cdot \frac{\partial a}{\partial x} + \frac{\partial v}{\partial b} \cdot \frac{\partial b}{\partial x} + \frac{\partial v}{\partial c} \cdot \frac{\partial c}{\partial x}$$

$$= \frac{\partial v}{\partial a} \cdot e^{x-y} + \frac{\partial v}{\partial b} \cdot e^{y-x} \cdot (-a) + \frac{\partial v}{\partial c} \cdot e^{z-x} \cdot (-1)$$

$$= e^{x-y} \frac{\partial v}{\partial a} + e^{y-x} (0) \frac{\partial v}{\partial b} - e^{z-x} \frac{\partial v}{\partial c}$$

$$\frac{\partial v}{\partial x} = e^{x-y} \frac{\partial v}{\partial a} - e^{z-x} \frac{\partial v}{\partial c} \quad \text{--- (1)}$$

$$\frac{\partial v}{\partial y} = -e^{x-y} \frac{\partial v}{\partial a} + e^{y-z} \frac{\partial v}{\partial b} \quad \text{--- (2)}$$

$$\frac{\partial v}{\partial z} = \frac{\partial v}{\partial a} (0) + \frac{\partial v}{\partial b} (e^{y-z}) + \frac{\partial v}{\partial c} (e^{z-x})$$

$$\frac{\partial v}{\partial z} = -e^{y-z} \frac{\partial v}{\partial b} + e^{z-x} \frac{\partial v}{\partial c} \quad \text{--- (3)}$$

Add equ. (1), (2), (3)

$$\therefore \frac{\partial v}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial v}{\partial z} = \left(e^{x-y} \frac{\partial v}{\partial a} - e^{z-x} \frac{\partial v}{\partial c} \right) +$$

$$\left(-e^{x-y} \frac{\partial v}{\partial a} + e^{y-z} \frac{\partial v}{\partial b} \right) + \left(-e^{y-z} \frac{\partial v}{\partial b} + e^{z-x} \frac{\partial v}{\partial c} \right)$$

$$\therefore \frac{\partial v}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial v}{\partial z} = 0$$

Hence proved, LHS = RHS.

Variable treated as constant :

If $x = u \tan v$, $y = u \sec v$ then we have 4 variables, x, y, u, v but here in a variable treated as constant we have to deal with only 3 variable at a time.

$\left(\frac{\partial u}{\partial y}\right)_x$ which can be read as, partial derivative of u w.r.t. y keeping x constant.

For this derivative, we require only 3 variable i.e. x, y, u . that means, we need to be eliminate 1 variable from given two equaⁿ.

Ex-1 if $u = 2x + 3y$, $v = 3x - 2y$, then p.t.

$$i) \left(\frac{\partial u}{\partial v}\right)_x \cdot \left(\frac{\partial v}{\partial y}\right)_u = \frac{13}{4}$$

$$ii) \left(\frac{\partial u}{\partial x}\right)_y \cdot \left(\frac{\partial x}{\partial v}\right)_u = \frac{4}{13}$$

also find the value of $\left(\frac{\partial u}{\partial x}\right)_y \cdot \left(\frac{\partial x}{\partial v}\right)_u$, $\left(\frac{\partial y}{\partial v}\right)_x \cdot \left(\frac{\partial v}{\partial y}\right)_u$

To find $\left(\frac{\partial u}{\partial x}\right)_y$, we have,

$$u = 2x + 3y \quad \text{--- ①} \quad v = 3x - 2y \quad \text{--- ②}$$

$$\therefore \left(\frac{\partial u}{\partial x}\right)_y = \frac{\partial (2x+3y)}{\partial x}$$

$$= 2 + 0$$

$$\left(\frac{\partial u}{\partial x}\right)_y = 2 \quad \text{--- (1)}$$

To find $\left(\frac{\partial v}{\partial y}\right)_u$

multiply equ. (1) by 3 & equ. (2) by 2

$$3u = 6x + 9y$$

$$2v = 6x - 4y$$

$$3u - 2v = 13y$$

$$3u - 13y = 2v$$

$$\therefore 3u - 13y = 2v$$

$$\therefore v = \frac{3u - 13y}{2}$$

diff. w.r.t. to y keeping u constant

$$\left(\frac{\partial v}{\partial y}\right)_u = \frac{\partial}{\partial y} \left(\frac{3u - 13y}{2} \right)_u$$

$$\therefore \left(\frac{\partial v}{\partial y}\right)_u = \frac{-13}{2} \quad \text{--- (4)}$$

To find $\left(\frac{\partial y}{\partial v}\right)_x$ we have,

$$v = 3x - 2y$$

$$V = 3x - 2y$$

$$V - 3x = -2y$$

$$\therefore y = \frac{3x - V}{2}$$

diff. w.r.t. to V keeping x constant

$$\therefore \left(\frac{\partial y}{\partial V}\right)_x = \frac{\partial}{\partial V} \left(\frac{3x - V}{2} \right)$$

$$\therefore \left(\frac{\partial y}{\partial V}\right)_x = \frac{-1}{2} \quad \text{--- (5)}$$

To find $\left(\frac{\partial x}{\partial U}\right)_V$

by using eqn (1) & (2) multiply by
② 2 to eqn (1) & 3 to eqn (2)

$$2U = 4x + 6y$$

$$3V = 9x - 6y$$

$$\text{+} \quad 2U + 3V = 13x$$

$$\therefore x = \frac{2U + 3V}{13}$$

$$\therefore \left(\frac{\partial x}{\partial U}\right)_V = \frac{\partial}{\partial U} \left(\frac{2U + 3V}{13} \right)_V$$

$$= \frac{1}{13} \cdot 2$$

$$\left(\frac{\partial x}{\partial U}\right)_V = \frac{2}{13}$$

--- (6)

from equ. (4) and (5)

$$\left(\frac{\partial y}{\partial v}\right)_x \cdot \left(\frac{\partial v}{\partial y}\right)_u = \frac{1}{2} \cdot \frac{13}{2} = \frac{13}{4}$$

from equ. (3) and (6)

$$\left(\frac{\partial u}{\partial x}\right)_y \cdot \left(\frac{\partial x}{\partial u}\right)_v = \frac{2}{13} \cdot \frac{2}{13} = \frac{4}{13}$$

Hence proved.

$\therefore \left(\frac{\partial u}{\partial x}\right)_v$ = To find.

multiply equ. (3) by 2 & equ. (2) by 3

$$2u = 4x + 6y$$

$$3v = 9x - 6y$$

$$\text{So } 2u + 3v = 13x$$

$$2u = 13x - 3v$$

$$u = \frac{13x - 3v}{2}$$

diff. w.r.t x keeping y, v const.

$$\left(\frac{\partial u}{\partial x}\right)_v = \frac{\partial}{\partial x} \left(\frac{13x - 3v}{2} \right)_v$$

$$= \frac{1}{2} \cdot 13$$

$$\left(\frac{\partial u}{\partial x}\right)_v = \frac{13}{2}$$

$$2 \times \frac{2}{13} \times \frac{1}{2} \times \frac{13}{2}$$

$$= 1$$

$\therefore$ The value of given expression is 1.

Unit
wwwimpEx-2 If $x = u \tan v$, $y = u \sec v$, then p.t.

$$\left(\frac{\partial u}{\partial x}\right)_y \cdot \left(\frac{\partial v}{\partial x}\right)_y = \left(\frac{\partial u}{\partial y}\right)_x \cdot \left(\frac{\partial v}{\partial y}\right)_x$$

→

$$x = u \tan v \quad \text{--- (1)} \quad , \quad y = u \sec v \quad \text{--- (2)}$$

$$\therefore \frac{x}{u} = \tan v \quad \text{--- (3)} \quad \quad \frac{y}{u} = \sec v \quad \text{--- (4)}$$

we know that

$$1 + \tan^2 v = \sec^2 v$$

$$\therefore \sec^2 v - \tan^2 v = 1$$

$$\frac{y^2}{u^2} - \frac{x^2}{u^2} = 1$$

$$\therefore \frac{y^2 - x^2}{u^2} = 1$$

$$\therefore y^2 - x^2 = u^2 \quad \text{--- (5)}$$

diff. w.r.t. x keeping y constant.

$$\left(\frac{\partial u^2}{\partial x}\right)_y = \left(\frac{\partial (y^2 - x^2)}{\partial x}\right)_y$$

$$2u \left(\frac{\partial u}{\partial x}\right)_y = 2y - 2x \quad \text{--- } y \text{ constant.}$$

$$\therefore 2u \left(\frac{\partial u}{\partial x}\right)_y = -2x$$

$$\therefore \left(\frac{\partial u}{\partial x}\right)_y = \frac{-2x}{2u} = \frac{-x}{u}$$

$$\therefore \left(\frac{\partial u}{\partial x}\right)_y = \frac{-x}{u}$$

diff. u w.r.t y keeping x constant.

$$\left(\frac{\partial u}{\partial y}\right)_x = \frac{\partial (y^2 - x^2)}{\partial y}$$

$$\left(\frac{\partial u^2}{\partial y}\right)_x = 2y$$

$$2u \left(\frac{\partial u}{\partial y}\right)_x = 2y$$

$$\therefore \left(\frac{\partial u}{\partial y}\right)_x = \frac{2y}{2u} = \frac{y}{u}$$

$$\therefore \left(\frac{\partial u}{\partial y}\right)_x = \frac{y}{u} \quad \text{--- (5)}$$

now, divide eqnⁿ (3) & (4).

$$\frac{\frac{x}{u}}{\frac{y}{u}} = \frac{\tan v}{\sec v}$$

$$\therefore \frac{x}{y} = \frac{\sin v}{\cos v} \cdot \frac{\cos v}{1}$$

$$\therefore \sin v = \frac{x}{y} \quad \text{--- (6)}$$

diff. w.r.t to x keeping y constant.

$$\left(\frac{\partial \sin v}{\partial x}\right)_y = \frac{\partial}{\partial x} \left(\frac{x}{y}\right)_y$$

$$\cos v \cdot \left(\frac{\partial v}{\partial x}\right)_y = \frac{1}{y}$$

$$\left(\frac{\partial v}{\partial x}\right)_y = \frac{1}{y \cos v} \quad \text{--- (7)}$$

diff. v w.r.t. y keeping x constant.

$$\left(\frac{\partial \sin v}{\partial y}\right)_x = \left(\frac{\partial x/y}{\partial y}\right)_x$$

$$\therefore \cos v \cdot \left(\frac{\partial v}{\partial y}\right)_x = \frac{-x}{y^2}$$

$$\therefore \left(\frac{\partial v}{\partial y}\right)_x = \frac{-x}{\cos v \cdot y^2} \quad \text{--- (8)}$$

from eqn. (5), (6), (7) and (8).

$$\left(\frac{\partial u}{\partial x}\right)_y \cdot \left(\frac{\partial v}{\partial x}\right)_y = \left(\frac{\partial u}{\partial y}\right)_x \cdot \left(\frac{\partial v}{\partial y}\right)_x$$

$$\frac{+x}{u} \cdot \frac{1}{y \cos v} = \frac{y}{u} \cdot \frac{-x}{\cos v \cdot y^2}$$

$$\tan v = \frac{\tan v}{y \cos v}$$

Hence proved.

Euler's Theorem :Homogeneous f^n :

A f^n of two variable x & y is said to be homogeneous of degree n , if it is possible to express

$$z_1 = x^n + (y/x)$$

$$z_2 = y^n + (x/y)$$

Euler's Theorem :

i) if z is a homogeneous f^n of two variable x & y of degree n , then

$$x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = n \cdot z$$

ii) if u is a homogeneous f^n of x, y of degree n and $z = f(u)$ then,

$$x \cdot \frac{\partial u}{\partial x} + y \cdot \frac{\partial u}{\partial y} = n \cdot \frac{f(u)}{f'(u)} \rightarrow g(u)$$

$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = g(u) [g'(u) - 1]$$

* Basic Identities :

1) $\sin^2 \theta + \cos^2 \theta = 1$

2) $1 + \tan^2 \theta = \sec^2 \theta$

3) $1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$

4) $\sin 2\theta = 2 \sin \theta \cos \theta$
 $= \frac{2 \tan \theta}{1 + \tan^2 \theta}$

5) $\cos 2\theta = \cos^2 \theta - \sin^2 \theta$
 $= 1 - 2 \sin^2 \theta$
 $= 2 \cos^2 \theta - 1$
 $= \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta}$

Ex-1 If $u = \tan^{-1} \left(\frac{x^3 + y^3}{x - y} \right)$ then p.t.

$$x^2 \cdot \frac{\partial^2 u}{\partial x^2} + 2xy \cdot \frac{\partial^2 u}{\partial x \partial y} + y^2 \cdot \frac{\partial^2 u}{\partial y^2} = (1 - 4 \sin^2 u) \sin 2u$$

$$u = \tan^{-1} \left(\frac{x^3 + y^3}{x - y} \right)$$

$$\therefore \tan u = \frac{x^3 + y^3}{x - y}$$

$$\tan u = \frac{x^3 \left(\frac{x^3}{x^3} + \frac{y^3}{x^3} \right)}{x \left(\frac{x}{x} + \frac{y}{x} \right)}$$

$$\therefore \tan u = x^2 \cdot \frac{\left(1 + \frac{y^3}{x^3}\right)}{(1 + y/x)}$$

$$\therefore \tan u = x^2 \cdot \left(\frac{1 + y^3/x^3}{1 + y/x} \right)$$

$$\therefore n = 2, \quad f(u) = \tan u$$

By Euler's Theorem,

$$x \cdot \frac{\partial u}{\partial x} + y \cdot \frac{\partial u}{\partial y} = n \cdot \frac{f(u)}{f'(u)} \Rightarrow g(u)$$

$$\therefore g(u) = n \frac{f(u)}{f'(u)}$$

$$= 2 \cdot \frac{\tan u}{\sec^2 u}$$

$$= 2 \cdot \frac{\sin u}{\cos u} \times \frac{\cos^2 u}{\cos 1}$$

$$g(u) = 2 \sin u \cdot \cos u$$

$$g(u) = \sin 2u$$

$$x^2 \cdot \frac{\partial^2 u}{\partial x^2} + 2xy \cdot \frac{\partial^2 u}{\partial x \partial y} + y^2 \cdot \frac{\partial^2 u}{\partial y^2} = g(u) \cdot [g'(u) - 1]$$

$$= \sin 2u [2 \cos 2u - 1]$$

$$= \sin 2u [2(1 - 2 \sin^2 u) - 1]$$

$$= \sin 2u [2 - 4 \sin^2 u - 1]$$

$$\text{RHS} = \sin 2u [1 - 4 \sin^2 u]$$

Hence proved.

Unit
VVImp
Ex.-2

$$u = \sin^{-1} \left(\frac{x+y}{\sqrt{x}+\sqrt{y}} \right) \quad \text{then P.T.}$$

$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = g(u) \cdot [g'(u)]^2$$

→

$$= \frac{1}{4} [\tan^3 u - \tan u]$$

$$u = \sin^{-1} \left(\frac{x+y}{\sqrt{x}+\sqrt{y}} \right)$$

$$\therefore \sin u = \frac{x^{1/2} (x/x + y/x)}{\sqrt{x} (\sqrt{x}/\sqrt{x} + \sqrt{y}/\sqrt{x})}$$

$$\therefore \sin u = \frac{\sqrt{x} \cdot (1 + y/x)}{(1 + \sqrt{y}/\sqrt{x})}$$

$$\sin u = \frac{\sqrt{x} \cdot (1 + y/x)}{(1 + \sqrt{y}/\sqrt{x})}$$

$$\therefore n = x^{1/2}, \quad f(u) = \sin u$$

$$\therefore x \cdot \frac{\partial u}{\partial x} + y \cdot \frac{\partial u}{\partial y} = \frac{f(u)}{f'(u)} \cdot n$$

$$g(u) = \frac{f(u)}{f'(u)} \cdot n$$

$$g(u) = \frac{1}{2} \cdot \frac{\sin u}{\cos u}$$

$$\therefore g(u) = \frac{\tan u}{2}$$

$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = g(u) [g(u) - 1]$$

$$= \frac{\tan u}{2} \left[\frac{\sec^2 u}{2} - 1 \right]$$

$$= \frac{\tan u}{2} \left[\frac{\sec^2 u - 2}{2} \right]$$

$$= \frac{\tan u}{4} [1 + \tan^2 u - 2]$$

$$= \frac{\tan u}{4} [\tan^2 u - 1]$$

$$= \frac{1}{4} [\tan^3 u - \tan u]$$

Hence proved.

Question bank's questions :

1. if $u = \sin^{-1}(y/x) + \sqrt{x^2 + y^2}$. Find the value of $x \cdot \frac{\partial u}{\partial x} + y \cdot \frac{\partial u}{\partial y}$

→

$$u = \sin^{-1}(y/x) + \sqrt{x^2 + y^2}$$

$$\text{Let } u = p + q$$

$$\therefore p = \sin^{-1}(y/x)$$

$$p = x^0 \cdot \sin^{-1}(y/x)$$

$$\therefore f(u) = p, \quad n = 0$$

By Euler's theorem,

$$x \cdot \frac{\partial p}{\partial x} + y \cdot \frac{\partial p}{\partial y} = n p$$

$$= 0 \cdot p$$

$$= 0 \quad \text{--- (1)}$$

now,

$$q = \sqrt{x^2 + y^2}$$

$$q = (x^2 + y^2)^{1/2}$$

$$q = (x^2)^{1/2} \cdot \left(\frac{x^2}{x^2} + \frac{y^2}{x^2} \right)^{1/2}$$

$$q = x \cdot \left(1 + \frac{y^2}{x^2} \right)^{1/2}$$

$$\therefore f(u) = q, \quad n = 1.$$

By Euler's theorem,

$$x \cdot \frac{\partial q}{\partial x} + y \cdot \frac{\partial q}{\partial y} = n \cdot q$$

$$= 1 \cdot \sqrt{x^2 + y^2}$$

$$= \sqrt{x^2 + y^2} \quad \text{--- (2)}$$

Add equ. (1) and (2).

$$\left(x \cdot \frac{\partial p}{\partial x} + y \cdot \frac{\partial p}{\partial y} \right) + \left(x \cdot \frac{\partial q}{\partial x} + y \cdot \frac{\partial q}{\partial y} \right)$$

$$= 0 + \sqrt{x^2 + y^2}$$

$$\underline{\text{Answer}} = \sqrt{x^2 + y^2}$$

2. if $u = \sin^{-1}(\sqrt{x^2+y^2})$, then find the value of $x^2 \cdot \frac{\partial^2 u}{\partial x^2} + 2xy \cdot \frac{\partial^2 u}{\partial x \partial y} + y^2 \cdot \frac{\partial^2 u}{\partial y^2}$

$$u = \sin^{-1}(\sqrt{x^2+y^2})$$

$$\therefore \sin u = (x^2+y^2)^{1/2}$$

$$\therefore \sin u = (x^2)^{1/2} \left(\frac{x^2}{x^2} + \frac{y^2}{x^2} \right)^{1/2}$$

$$\sin u = x \cdot \left(1 + \frac{y^2}{x^2} \right)^{1/2}$$

$$\therefore \underline{n=1} \quad f(u) = \sin u$$

$$\therefore x^2 \cdot \frac{\partial^2 u}{\partial x^2} + 2xy \cdot \frac{\partial^2 u}{\partial x \partial y} + y^2 \cdot \frac{\partial^2 u}{\partial y^2} = n \cdot \frac{f(u)}{f'(u)}$$

$$g(u) = 1 \cdot \frac{\sin u}{\cos u}$$

$$g(u) = \tan u$$

$$= g(u) (g'(u) - 1)$$

$$= \tan u (\sec^2 u - 1)$$

$$= \tan u (1 + \tan^2 u - 1)$$

$$= \tan^3 u$$

$$\therefore \underline{= \tan^3 u}$$

3. if $u = \operatorname{cosec}^{-1} \sqrt{\frac{x^{1/2} + y^{1/2}}{x^{1/3} + y^{1/3}}}$ s.t.

$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = \frac{\tan u}{12} \left(\frac{13}{12} + \frac{\tan^2 u}{12} \right)$$

$$\operatorname{cosec} u = \left(\frac{x^{1/2} + y^{1/2}}{x^{1/3} + y^{1/3}} \right)^{1/2}$$

$$\therefore \operatorname{cosec} u = \frac{(x^{1/2})^{1/2} \left(1 + (y/x)^{1/2} \right)^{1/2}}{(x^{1/3})^{1/2} \left(1 + (y/x)^{1/3} \right)}$$

$$\therefore \operatorname{cosec} u = \frac{x^{1/4}}{x^{1/6}} \times \frac{\left(1 + (y/x)^{1/2} \right)^{1/2}}{\left(1 + (y/x)^{1/3} \right)}$$

$$\therefore f(u) = \operatorname{cosec} u, \quad n = \frac{1}{12}$$

$$\therefore g(u) = \frac{n \cdot f(u)}{f(u)} = \frac{1}{12} \times \frac{\operatorname{cosec} u}{-\operatorname{cosec} u \cdot \cot u}$$

$$\therefore g(u) = \frac{1}{12} \times \frac{\operatorname{cosec} u}{-\operatorname{cosec} u \cdot \cot u}$$

$$\therefore g(u) = -\frac{1}{12} \frac{1}{\cot u} = -\frac{1}{12} \tan u$$

$$= -\frac{1}{12} \tan u \left(-\frac{\sec^2 u}{12} - 1 \right) = -\frac{1}{12} \tan u \left(-\frac{1}{12} - \frac{\tan^2 u}{12} - 1 \right)$$

$$= +\frac{1}{12 \times 12} \tan u \left(1 + \tan^2 u - 1 \right)$$

$$= \frac{1}{144} \tan u \cdot \tan^2 u = \frac{\tan^3 u}{144}$$

$$= \frac{-1}{12} \tan U \left[-\frac{13}{12} - \frac{\tan^2 U}{12} \right]$$

multiply by (-)

$$\text{RHS} = \frac{1}{12} \tan U \left[\frac{13}{12} + \frac{\tan^2 U}{12} \right]$$

$$\therefore \text{LHS} = \text{RHS}$$

Hence proved.

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